

Before proving Theorem 4, we look at several examples to gain a better understanding of what it says.

EXAMPLE 3 Applying the Fundamental Theorem

Use the Fundamental Theorem to find

- (a) $\frac{d}{dx} \int_a^x \cos t \, dt$
- (b) $\frac{d}{dx} \int_0^x \frac{1}{1+t^2} \, dt$
- (c) $\frac{dy}{dx}$ if $y = \int_x^5 3t \sin t \, dt$
- (d) $\frac{dy}{dx}$ if $y = \int_1^{x^2} \cos t \, dt$

Solution

$$(a) \quad \frac{d}{dx} \int_a^x \cos t \, dt = \cos x \quad \text{Eq. 2 with } f(t) = \cos t$$

$$(b) \quad \frac{d}{dx} \int_0^x \frac{1}{1+t^2} \, dt = \frac{1}{1+x^2} \quad \text{Eq. 2 with } f(t) = \frac{1}{1+t^2}$$

(c) Rule 1 for integrals in Table 5.3 of Section 5.3 sets this up for the Fundamental Theorem.

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \int_x^5 3t \sin t \, dt = \frac{d}{dx} \left(- \int_5^x 3t \sin t \, dt \right) && \text{Rule 1} \\ &= - \frac{d}{dx} \int_5^x 3t \sin t \, dt \\ &= -3x \sin x \end{aligned}$$

(d) The upper limit of integration is not x but x^2 . This makes y a composite of the two functions,

$$y = \int_1^u \cos t \, dt \quad \text{and} \quad u = x^2.$$

We must therefore apply the Chain Rule when finding dy/dx .

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \left(\frac{d}{du} \int_1^u \cos t \, dt \right) \cdot \frac{du}{dx} \\ &= \cos u \cdot \frac{du}{dx} \\ &= \cos(x^2) \cdot 2x \\ &= 2x \cos x^2 \end{aligned}$$



EXAMPLE 4 Constructing a Function with a Given Derivative and Value

Find a function $y = f(x)$ on the domain $(-\pi/2, \pi/2)$ with derivative

$$\frac{dy}{dx} = \tan x$$

that satisfies the condition $f(3) = 5$.

Solution The Fundamental Theorem makes it easy to construct a function with derivative $\tan x$ that equals 0 at $x = 3$:

$$y = \int_3^x \tan t \, dt.$$

Since $y(3) = \int_3^3 \tan t \, dt = 0$, we have only to add 5 to this function to construct one with derivative $\tan x$ whose value at $x = 3$ is 5:

$$f(x) = \int_3^x \tan t \, dt + 5. \quad \blacksquare$$

Although the solution to the problem in Example 4 satisfies the two required conditions, you might ask whether it is in a useful form. The answer is yes, since today we have computers and calculators that are capable of approximating integrals. In Chapter 7 we will learn to write the solution in Example 4 exactly as

$$y = \ln \left| \frac{\cos 3}{\cos x} \right| + 5.$$

We now give a proof of the Fundamental Theorem for an arbitrary continuous function.

Proof of Theorem 4 We prove the Fundamental Theorem by applying the definition of the derivative directly to the function $F(x)$, when x and $x + h$ are in (a, b) . This means writing out the difference quotient

$$\frac{F(x + h) - F(x)}{h} \quad (3)$$

and showing that its limit as $h \rightarrow 0$ is the number $f(x)$ for each x in (a, b) .

When we replace $F(x + h)$ and $F(x)$ by their defining integrals, the numerator in Equation (3) becomes

$$F(x + h) - F(x) = \int_a^{x+h} f(t) \, dt - \int_a^x f(t) \, dt.$$

The Additivity Rule for integrals (Table 5.3, Rule 5) simplifies the right side to

$$\int_x^{x+h} f(t) \, dt,$$

so that Equation (3) becomes

$$\begin{aligned} \frac{F(x + h) - F(x)}{h} &= \frac{1}{h} [F(x + h) - F(x)] \\ &= \frac{1}{h} \int_x^{x+h} f(t) \, dt. \end{aligned} \quad (4)$$

According to the Mean Value Theorem for Definite Integrals, the value of the last expression in Equation (4) is one of the values taken on by f in the interval between x and $x + h$. That is, for some number c in this interval,

$$\frac{1}{h} \int_x^{x+h} f(t) dt = f(c). \quad (5)$$

As $h \rightarrow 0$, $x + h$ approaches x , forcing c to approach x also (because c is trapped between x and $x + h$). Since f is continuous at x , $f(c)$ approaches $f(x)$:

$$\lim_{h \rightarrow 0} f(c) = f(x). \quad (6)$$

Going back to the beginning, then, we have

$$\begin{aligned} \frac{dF}{dx} &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} && \text{Definition of derivative} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt && \text{Eq. (4)} \\ &= \lim_{h \rightarrow 0} f(c) && \text{Eq. (5)} \\ &= f(x). && \text{Eq. (6)} \end{aligned}$$

If $x = a$ or b , then the limit of Equation (3) is interpreted as a one-sided limit with $h \rightarrow 0^+$ or $h \rightarrow 0^-$, respectively. Then Theorem 1 in Section 3.1 shows that F is continuous for every point of $[a, b]$. This concludes the proof. ■

Fundamental Theorem, Part 2 (The Evaluation Theorem)

We now come to the second part of the Fundamental Theorem of Calculus. This part describes how to evaluate definite integrals without having to calculate limits of Riemann sums. Instead we find and evaluate an antiderivative at the upper and lower limits of integration.

THEOREM 4 (Continued) The Fundamental Theorem of Calculus Part 2

If f is continuous at every point of $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Proof Part 1 of the Fundamental Theorem tells us that an antiderivative of f exists, namely

$$G(x) = \int_a^x f(t) dt.$$

Thus, if F is *any* antiderivative of f , then $F(x) = G(x) + C$ for some constant C for $a < x < b$ (by Corollary 2 of the Mean Value Theorem for Derivatives, Section 4.2). Since both F and G are continuous on $[a, b]$, we see that $F(x) = G(x) + C$ also holds when $x = a$ and $x = b$ by taking one-sided limits (as $x \rightarrow a^+$ and $x \rightarrow b^-$).

Evaluating $F(b) - F(a)$, we have

$$\begin{aligned}
 F(b) - F(a) &= [G(b) + C] - [G(a) + C] \\
 &= G(b) - G(a) \\
 &= \int_a^b f(t) dt - \int_a^a f(t) dt \\
 &= \int_a^b f(t) dt - 0 \\
 &= \int_a^b f(t) dt.
 \end{aligned}$$

The theorem says that to calculate the definite integral of f over $[a, b]$ all we need to do is:

1. Find an antiderivative F of f , and
2. Calculate the number $\int_a^b f(x) dx = F(b) - F(a)$.

The usual notation for $F(b) - F(a)$ is

$$F(x) \Big|_a^b \quad \text{or} \quad \left[F(x) \right]_a^b,$$

depending on whether F has one or more terms.

EXAMPLE 5 Evaluating Integrals

- (a) $\int_0^\pi \cos x dx = \sin x \Big|_0^\pi = \sin \pi - \sin 0 = 0 - 0 = 0$
- (b) $\int_{-\pi/4}^0 \sec x \tan x dx = \sec x \Big|_{-\pi/4}^0 = \sec 0 - \sec \left(-\frac{\pi}{4}\right) = 1 - \sqrt{2}$
- (c) $\int_1^4 \left(\frac{3}{2} \sqrt{x} - \frac{4}{x^2} \right) dx = \left[x^{3/2} + \frac{4}{x} \right]_1^4$
 $= \left[(4)^{3/2} + \frac{4}{4} \right] - \left[(1)^{3/2} + \frac{4}{1} \right]$
 $= [8 + 1] - [5] = 4.$

The process used in Example 5 was much easier than a Riemann sum computation.

The conclusions of the Fundamental Theorem tell us several things. Equation (2) can be rewritten as

$$\frac{d}{dx} \int_a^x f(t) dt = \frac{dF}{dx} = f(x),$$

which says that if you first integrate the function f and then differentiate the result, you get the function f back again. Likewise, the equation

$$\int_a^x \frac{dF}{dt} dt = \int_a^x f(t) dt = F(x) - F(a)$$

says that if you first differentiate the function F and then integrate the result, you get the function F back (adjusted by an integration constant). In a sense, the processes of integra-

tion and differentiation are “inverses” of each other. The Fundamental Theorem also says that every continuous function f has an antiderivative F . And it says that the differential equation $dy/dx = f(x)$ has a solution (namely, the function $y = F(x)$) for every continuous function f .

Total Area

The Riemann sum contains terms such as $f(c_k) \Delta_k$ which give the area of a rectangle when $f(c_k)$ is positive. When $f(c_k)$ is negative, then the product $f(c_k) \Delta_k$ is the negative of the rectangle's area. When we add up such terms for a negative function we get the negative of the area between the curve and the x -axis. If we then take the absolute value, we obtain the correct positive area.

EXAMPLE 6 Finding Area Using Antiderivatives

Calculate the area bounded by the x -axis and the parabola $y = 6 - x - x^2$.

Solution We find where the curve crosses the x -axis by setting

$$y = 0 = 6 - x - x^2 = (3 + x)(2 - x),$$

which gives

$$x = -3 \quad \text{or} \quad x = 2.$$

The curve is sketched in Figure 5.21, and is nonnegative on $[-3, 2]$.

The area is

$$\begin{aligned} \int_{-3}^2 (6 - x - x^2) dx &= \left[6x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-3}^2 \\ &= \left(12 - 2 - \frac{8}{3} \right) - \left(-18 - \frac{9}{2} + \frac{27}{3} \right) = 20\frac{5}{6}. \end{aligned}$$

The curve in Figure 5.21 is an arch of a parabola, and it is interesting to note that the area under such an arch is exactly equal to two-thirds the base times the altitude:

$$\frac{2}{3}(5)\left(\frac{25}{4}\right) = \frac{125}{6} = 20\frac{5}{6}. \quad \blacksquare$$

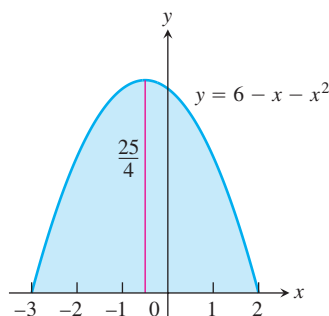


FIGURE 5.21 The area of this parabolic arch is calculated with a definite integral (Example 6).

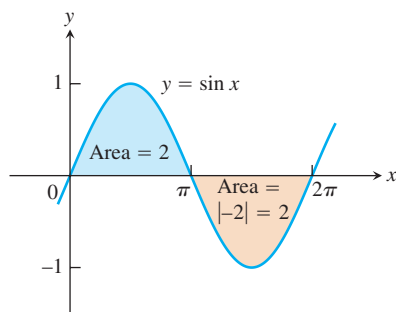


FIGURE 5.22 The total area between $y = \sin x$ and the x -axis for $0 \leq x \leq 2\pi$ is the sum of the absolute values of two integrals (Example 7).

To compute the area of the region bounded by the graph of a function $y = f(x)$ and the x -axis requires more care when the function takes on both positive and negative values. We must be careful to break up the interval $[a, b]$ into subintervals on which the function doesn't change sign. Otherwise we might get cancellation between positive and negative signed areas, leading to an incorrect total. The correct total area is obtained by adding the absolute value of the definite integral over each subinterval where $f(x)$ does not change sign. The term “area” will be taken to mean *total area*.

EXAMPLE 7 Canceling Areas

Figure 5.22 shows the graph of the function $f(x) = \sin x$ between $x = 0$ and $x = 2\pi$. Compute

- the definite integral of $f(x)$ over $[0, 2\pi]$.
- the area between the graph of $f(x)$ and the x -axis over $[0, 2\pi]$.

Solution The definite integral for $f(x) = \sin x$ is given by

$$\int_0^{2\pi} \sin x \, dx = -\cos x \Big|_0^{2\pi} = -[\cos 2\pi - \cos 0] = -[1 - 1] = 0.$$

The definite integral is zero because the portions of the graph above and below the x -axis make canceling contributions.

The area between the graph of $f(x)$ and the x -axis over $[0, 2\pi]$ is calculated by breaking up the domain of $\sin x$ into two pieces: the interval $[0, \pi]$ over which it is nonnegative and the interval $[\pi, 2\pi]$ over which it is nonpositive.

$$\begin{aligned} \int_0^{\pi} \sin x \, dx &= -\cos x \Big|_0^{\pi} = -[\cos \pi - \cos 0] = -[-1 - 1] = 2. \\ \int_{\pi}^{2\pi} \sin x \, dx &= -\cos x \Big|_{\pi}^{2\pi} = -[\cos 2\pi - \cos \pi] = -[1 - (-1)] = -2. \end{aligned}$$

The second integral gives a negative value. The area between the graph and the axis is obtained by adding the absolute values

$$\text{Area} = |2| + |-2| = 4. \quad \blacksquare$$

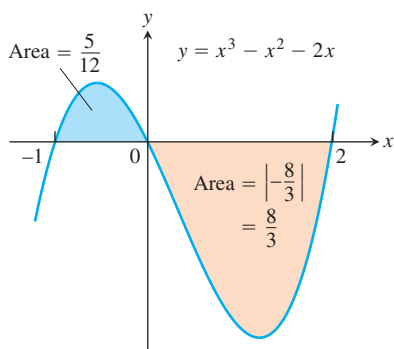


FIGURE 5.23 The region between the curve $y = x^3 - x^2 - 2x$ and the x -axis (Example 8).

Summary:

To find the area between the graph of $y = f(x)$ and the x -axis over the interval $[a, b]$, do the following:

1. Subdivide $[a, b]$ at the zeros of f .
2. Integrate f over each subinterval.
3. Add the absolute values of the integrals.

EXAMPLE 8 Finding Area Using Antiderivatives

Find the area of the region between the x -axis and the graph of $f(x) = x^3 - x^2 - 2x$, $-1 \leq x \leq 2$.

Solution First find the zeros of f . Since

$$f(x) = x^3 - x^2 - 2x = x(x^2 - x - 2) = x(x + 1)(x - 2),$$

the zeros are $x = 0, -1$, and 2 (Figure 5.23). The zeros subdivide $[-1, 2]$ into two subintervals: $[-1, 0]$, on which $f \geq 0$, and $[0, 2]$, on which $f \leq 0$. We integrate f over each subinterval and add the absolute values of the calculated integrals.

$$\begin{aligned} \int_{-1}^0 (x^3 - x^2 - 2x) \, dx &= \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 = 0 - \left[\frac{1}{4} + \frac{1}{3} - 1 \right] = \frac{5}{12} \\ \int_0^2 (x^3 - x^2 - 2x) \, dx &= \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2 = \left[4 - \frac{8}{3} - 4 \right] - 0 = -\frac{8}{3} \end{aligned}$$

The total enclosed area is obtained by adding the absolute values of the calculated integrals,

$$\text{Total enclosed area} = \frac{5}{12} + \left| -\frac{8}{3} \right| = \frac{37}{12}. \quad \blacksquare$$

EXERCISES 5.4

Evaluating Integrals

Evaluate the integrals in Exercises 1–26.

1. $\int_{-2}^0 (2x + 5) dx$
2. $\int_{-3}^4 \left(5 - \frac{x}{2}\right) dx$
3. $\int_0^4 \left(3x - \frac{x^3}{4}\right) dx$
4. $\int_{-2}^2 (x^3 - 2x + 3) dx$
5. $\int_0^1 (x^2 + \sqrt{x}) dx$
6. $\int_0^5 x^{3/2} dx$
7. $\int_1^{32} x^{-6/5} dx$
8. $\int_{-2}^{-1} \frac{2}{x^2} dx$
9. $\int_0^{\pi} \sin x dx$
10. $\int_0^{\pi} (1 + \cos x) dx$
11. $\int_0^{\pi/3} 2 \sec^2 x dx$
12. $\int_{\pi/6}^{5\pi/6} \csc^2 x dx$
13. $\int_{\pi/4}^{3\pi/4} \csc \theta \cot \theta d\theta$
14. $\int_0^{\pi/3} 4 \sec u \tan u du$
15. $\int_{\pi/2}^0 \frac{1 + \cos 2t}{2} dt$
16. $\int_{-\pi/3}^{\pi/3} \frac{1 - \cos 2t}{2} dt$
17. $\int_{-\pi/2}^{\pi/2} (8y^2 + \sin y) dy$
18. $\int_{-\pi/3}^{-\pi/4} \left(4 \sec^2 t + \frac{\pi}{t^2}\right) dt$
19. $\int_1^{-1} (r + 1)^2 dr$
20. $\int_{-\sqrt{3}}^{\sqrt{3}} (t + 1)(t^2 + 4) dt$
21. $\int_{\sqrt{2}}^1 \left(\frac{u^7}{2} - \frac{1}{u^5}\right) du$
22. $\int_{1/2}^1 \left(\frac{1}{v^3} - \frac{1}{v^4}\right) dv$
23. $\int_1^{\sqrt{2}} \frac{s^2 + \sqrt{s}}{s^2} ds$
24. $\int_9^4 \frac{1 - \sqrt{u}}{\sqrt{u}} du$
25. $\int_{-4}^4 |x| dx$
26. $\int_0^{\pi} \frac{1}{2} (\cos x + |\cos x|) dx$

Derivatives of Integrals

Find the derivatives in Exercises 27–30.

- a. by evaluating the integral and differentiating the result.
- b. by differentiating the integral directly.

27. $\frac{d}{dx} \int_0^{\sqrt{x}} \cos t dt$
28. $\frac{d}{dx} \int_1^{\sin x} 3t^2 dt$
29. $\frac{d}{dt} \int_0^{t^4} \sqrt{u} du$
30. $\frac{d}{d\theta} \int_0^{\tan \theta} \sec^2 y dy$

Find dy/dx in Exercises 31–36.

31. $y = \int_0^x \sqrt{1 + t^2} dt$
32. $y = \int_1^x \frac{1}{t} dt, \quad x > 0$
33. $y = \int_{\sqrt{x}}^0 \sin(t^2) dt$
34. $y = \int_0^{x^2} \cos \sqrt{t} dt$

$$35. y = \int_0^{\sin x} \frac{dt}{\sqrt{1 - t^2}}, \quad |x| < \frac{\pi}{2}$$

$$36. y = \int_{\tan x}^0 \frac{dt}{1 + t^2}$$

Area

In Exercises 37–42, find the total area between the region and the x -axis.

$$37. y = -x^2 - 2x, \quad -3 \leq x \leq 2$$

$$38. y = 3x^2 - 3, \quad -2 \leq x \leq 2$$

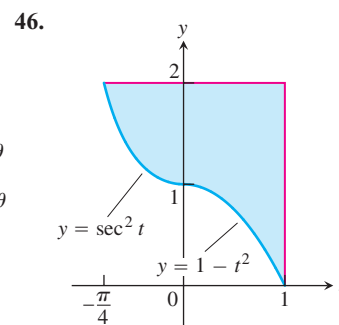
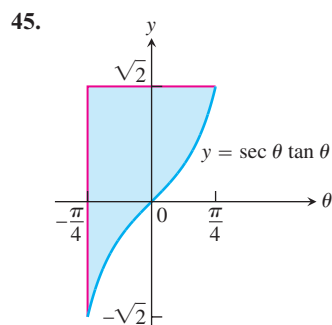
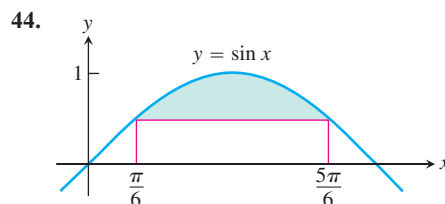
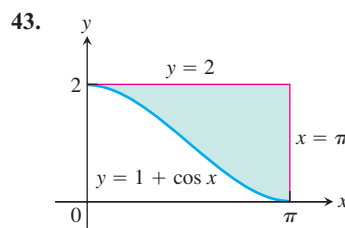
$$39. y = x^3 - 3x^2 + 2x, \quad 0 \leq x \leq 2$$

$$40. y = x^3 - 4x, \quad -2 \leq x \leq 2$$

$$41. y = x^{1/3}, \quad -1 \leq x \leq 8$$

$$42. y = x^{1/3} - x, \quad -1 \leq x \leq 8$$

Find the areas of the shaded regions in Exercises 43–46.



Initial Value Problems

Each of the following functions solves one of the initial value problems in Exercises 47–50. Which function solves which problem? Give brief reasons for your answers.

- a. $y = \int_1^x \frac{1}{t} dt - 3$ b. $y = \int_0^x \sec t \, dt + 4$
 c. $y = \int_{-1}^x \sec t \, dt + 4$ d. $y = \int_{\pi}^x \frac{1}{t} dt - 3$
 47. $\frac{dy}{dx} = \frac{1}{x}$, $y(\pi) = -3$ 48. $y' = \sec x$, $y(-1) = 4$
 49. $y' = \sec x$, $y(0) = 4$ 50. $y' = \frac{1}{x}$, $y(1) = -3$

Express the solutions of the initial value problems in Exercises 51–54 in terms of integrals.

51. $\frac{dy}{dx} = \sec x$, $y(2) = 3$
 52. $\frac{dy}{dx} = \sqrt{1+x^2}$, $y(1) = -2$
 53. $\frac{ds}{dt} = f(t)$, $s(t_0) = s_0$
 54. $\frac{dv}{dt} = g(t)$, $v(t_0) = v_0$

Applications

55. **Archimedes' area formula for parabolas** Archimedes (287–212 B.C.), inventor, military engineer, physicist, and the greatest mathematician of classical times in the Western world, discovered that the area under a parabolic arch is two-thirds the base times the height. Sketch the parabolic arch $y = h - (4h/b^2)x^2$, $-b/2 \leq x \leq b/2$, assuming that h and b are positive. Then use calculus to find the area of the region enclosed between the arch and the x -axis.
56. **Revenue from marginal revenue** Suppose that a company's marginal revenue from the manufacture and sale of egg beaters is

$$\frac{dr}{dx} = 2 - 2/(x+1)^2,$$

where r is measured in thousands of dollars and x in thousands of units. How much money should the company expect from a production run of $x = 3$ thousand egg beaters? To find out, integrate the marginal revenue from $x = 0$ to $x = 3$.

57. **Cost from marginal cost** The marginal cost of printing a poster when x posters have been printed is

$$\frac{dc}{dx} = \frac{1}{2\sqrt{x}}$$

dollars. Find $c(100) - c(1)$, the cost of printing posters 2–100.

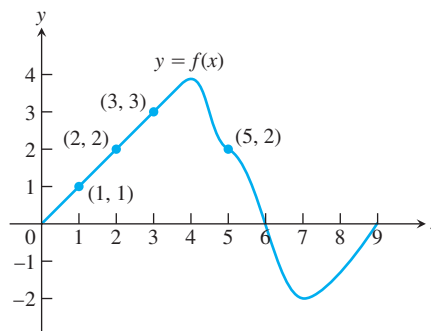
58. (Continuation of Exercise 57.) Find $c(400) - c(100)$, the cost of printing posters 101–400.

Drawing Conclusions About Motion from Graphs

59. Suppose that f is the differentiable function shown in the accompanying graph and that the position at time t (sec) of a particle moving along a coordinate axis is

$$s = \int_0^t f(x) \, dx$$

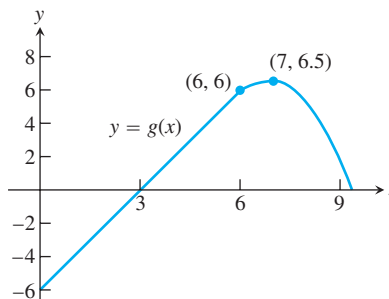
meters. Use the graph to answer the following questions. Give reasons for your answers.



- What is the particle's velocity at time $t = 5$?
 - Is the acceleration of the particle at time $t = 5$ positive, or negative?
 - What is the particle's position at time $t = 3$?
 - At what time during the first 9 sec does s have its largest value?
 - Approximately when is the acceleration zero?
 - When is the particle moving toward the origin? away from the origin?
 - On which side of the origin does the particle lie at time $t = 9$?
60. Suppose that g is the differentiable function graphed here and that the position at time t (sec) of a particle moving along a coordinate axis is

$$s = \int_0^t g(x) \, dx$$

meters. Use the graph to answer the following questions. Give reasons for your answers.



- What is the particle's velocity at $t = 3$?
- Is the acceleration at time $t = 3$ positive, or negative?
- What is the particle's position at time $t = 3$?
- When does the particle pass through the origin?
- When is the acceleration zero?
- When is the particle moving away from the origin? toward the origin?
- On which side of the origin does the particle lie at $t = 9$?

Theory and Examples

- Show that if k is a positive constant, then the area between the x -axis and one arch of the curve $y = \sin kx$ is $2/k$.
- Find

$$\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t^2}{t^4 + 1} dt.$$

- Suppose $\int_1^x f(t) dt = x^2 - 2x + 1$. Find $f(x)$.
- Find $f(4)$ if $\int_0^x f(t) dt = x \cos \pi x$.
- Find the linearization of

$$f(x) = 2 - \int_2^{x+1} \frac{9}{1+t} dt$$

at $x = 1$.

- Find the linearization of

$$g(x) = 3 + \int_1^{x^2} \sec(t-1) dt$$

at $x = -1$.

- Suppose that f has a positive derivative for all values of x and that $f(1) = 0$. Which of the following statements must be true of the function

$$g(x) = \int_0^x f(t) dt?$$

Give reasons for your answers.

- g is a differentiable function of x .
 - g is a continuous function of x .
 - The graph of g has a horizontal tangent at $x = 1$.
 - g has a local maximum at $x = 1$.
 - g has a local minimum at $x = 1$.
 - The graph of g has an inflection point at $x = 1$.
 - The graph of dg/dx crosses the x -axis at $x = 1$.
- Suppose that f has a negative derivative for all values of x and that $f(1) = 0$. Which of the following statements must be true of the function

$$h(x) = \int_0^x f(t) dt?$$

Give reasons for your answers.

- h is a twice-differentiable function of x .
- h and dh/dx are both continuous.
- The graph of h has a horizontal tangent at $x = 1$.
- h has a local maximum at $x = 1$.
- h has a local minimum at $x = 1$.
- The graph of h has an inflection point at $x = 1$.
- The graph of dh/dx crosses the x -axis at $x = 1$.

T 69. The Fundamental Theorem If f is continuous, we expect

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt$$

to equal $f(x)$, as in the proof of Part 1 of the Fundamental Theorem. For instance, if $f(t) = \cos t$, then

$$\frac{1}{h} \int_x^{x+h} \cos t dt = \frac{\sin(x+h) - \sin x}{h}. \quad (7)$$

The right-hand side of Equation (7) is the difference quotient for the derivative of the sine, and we expect its limit as $h \rightarrow 0$ to be $\cos x$.

Graph $\cos x$ for $-\pi \leq x \leq 2\pi$. Then, in a different color if possible, graph the right-hand side of Equation (7) as a function of x for $h = 2, 1, 0.5$, and 0.1 . Watch how the latter curves converge to the graph of the cosine as $h \rightarrow 0$.

T 70. Repeat Exercise 69 for $f(t) = 3t^2$. What is

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} 3t^2 dt = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}?$$

Graph $f(x) = 3x^2$ for $-1 \leq x \leq 1$. Then graph the quotient $((x+h)^3 - x^3)/h$ as a function of x for $h = 1, 0.5, 0.2$, and 0.1 . Watch how the latter curves converge to the graph of $3x^2$ as $h \rightarrow 0$.

COMPUTER EXPLORATIONS

In Exercises 71–74, let $F(x) = \int_a^x f(t) dt$ for the specified function f and interval $[a, b]$. Use a CAS to perform the following steps and answer the questions posed.

- Plot the functions f and F together over $[a, b]$.
- Solve the equation $F'(x) = 0$. What can you see to be true about the graphs of f and F at points where $F'(x) = 0$? Is your observation borne out by Part 1 of the Fundamental Theorem coupled with information provided by the first derivative? Explain your answer.
- Over what intervals (approximately) is the function F increasing and decreasing? What is true about f over those intervals?
- Calculate the derivative f' and plot it together with F . What can you see to be true about the graph of F at points where $f'(x) = 0$? Is your observation borne out by Part 1 of the Fundamental Theorem? Explain your answer.

71. $f(x) = x^3 - 4x^2 + 3x, \quad [0, 4]$

72. $f(x) = 2x^4 - 17x^3 + 46x^2 - 43x + 12, \quad \left[0, \frac{9}{2}\right]$

73. $f(x) = \sin 2x \cos \frac{x}{3}, \quad [0, 2\pi]$

74. $f(x) = x \cos \pi x, \quad [0, 2\pi]$

In Exercises 75–78, let $F(x) = \int_a^{u(x)} f(t) dt$ for the specified a , u , and f . Use a CAS to perform the following steps and answer the questions posed.

- Find the domain of F .
- Calculate $F'(x)$ and determine its zeros. For what points in its domain is F increasing? decreasing?
- Calculate $F''(x)$ and determine its zero. Identify the local extrema and the points of inflection of F .

- Using the information from parts (a)–(c), draw a rough hand-sketch of $y = F(x)$ over its domain. Then graph $F(x)$ on your CAS to support your sketch.

75. $a = 1, \quad u(x) = x^2, \quad f(x) = \sqrt{1 - x^2}$

76. $a = 0, \quad u(x) = x^2, \quad f(x) = \sqrt{1 - x^2}$

77. $a = 0, \quad u(x) = 1 - x, \quad f(x) = x^2 - 2x - 3$

78. $a = 0, \quad u(x) = 1 - x^2, \quad f(x) = x^2 - 2x - 3$

In Exercises 79 and 80, assume that f is continuous and $u(x)$ is twice-differentiable.

79. Calculate $\frac{d}{dx} \int_a^{u(x)} f(t) dt$ and check your answer using a CAS.

80. Calculate $\frac{d^2}{dx^2} \int_a^{u(x)} f(t) dt$ and check your answer using a CAS.

5.5

Indefinite Integrals and the Substitution Rule

A definite integral is a number defined by taking the limit of Riemann sums associated with partitions of a finite closed interval whose norms go to zero. The Fundamental Theorem of Calculus says that a definite integral of a continuous function can be computed easily if we can find an antiderivative of the function. Antiderivatives generally turn out to be more difficult to find than derivatives. However, it is well worth the effort to learn techniques for computing them.

Recall from Section 4.8 that the set of *all* antiderivatives of the function f is called the **indefinite integral** of f with respect to x , and is symbolized by

$$\int f(x) \, dx.$$

The connection between antiderivatives and the definite integral stated in the Fundamental Theorem now explains this notation. When finding the indefinite integral of a function f , remember that it always includes an arbitrary constant C .

We must distinguish carefully between definite and indefinite integrals. A definite integral $\int_a^b f(x) \, dx$ is a *number*. An indefinite integral $\int f(x) \, dx$ is a *function* plus an arbitrary constant C .

So far, we have only been able to find antiderivatives of functions that are clearly recognizable as derivatives. In this section we begin to develop more general techniques for finding antiderivatives. The first integration techniques we develop are obtained by inverting rules for finding derivatives, such as the Power Rule and the Chain Rule.

The Power Rule in Integral Form

If u is a differentiable function of x and n is a rational number different from -1 , the Chain Rule tells us that

$$\frac{d}{dx} \left(\frac{u^{n+1}}{n+1} \right) = u^n \frac{du}{dx}.$$

From another point of view, this same equation says that $u^{n+1}/(n+1)$ is one of the anti-derivatives of the function $u^n(du/dx)$. Therefore,

$$\int \left(u^n \frac{du}{dx} \right) dx = \frac{u^{n+1}}{n+1} + C.$$

The integral on the left-hand side of this equation is usually written in the simpler “differential” form,

$$\int u^n du,$$

obtained by treating the dx ’s as differentials that cancel. We are thus led to the following rule.

If u is any differentiable function, then

$$\int u^n du = \frac{u^{n+1}}{n+1} + C \quad (n \neq -1, n \text{ rational}). \quad (1)$$

Equation (1) actually holds for any real exponent $n \neq -1$, as we see in Chapter 7.

In deriving Equation (1), we assumed u to be a differentiable function of the variable x , but the name of the variable does not matter and does not appear in the final formula. We could have represented the variable with θ , t , y , or any other letter. Equation (1) says that whenever we can cast an integral in the form

$$\int u^n du, \quad (n \neq -1),$$

with u a differentiable function and du its differential, we can evaluate the integral as $[u^{n+1}/(n+1)] + C$.

EXAMPLE 1 Using the Power Rule

$$\begin{aligned} \int \sqrt{1+y^2} \cdot 2y \, dy &= \int \sqrt{u} \cdot \left(\frac{du}{dy} \right) dy && \text{Let } u = 1 + y^2, \\ &= \int u^{1/2} du && du/dy = 2y. \\ &= \frac{u^{(1/2)+1}}{(1/2)+1} + C && \text{Integrate, using Eq. (1) with } n = 1/2. \\ &= \frac{2}{3} u^{3/2} + C && \text{Simpler form} \\ &= \frac{2}{3} (1 + y^2)^{3/2} + C && \text{Replace } u \text{ by } 1 + y^2. \quad \blacksquare \end{aligned}$$

EXAMPLE 2 Adjusting the Integrand by a Constant

$$\begin{aligned}
 \int \sqrt{4t-1} \, dt &= \int \frac{1}{4} \cdot \sqrt{4t-1} \cdot 4 \, dt \\
 &= \frac{1}{4} \int \sqrt{u} \cdot \left(\frac{du}{dt} \right) dt && \text{Let } u = 4t - 1, \\
 &&& du/dt = 4. \\
 &= \frac{1}{4} \int u^{1/2} \, du && \text{With the } 1/4 \text{ out front,} \\
 &&& \text{the integral is now in} \\
 &&& \text{standard form.} \\
 &= \frac{1}{4} \cdot \frac{u^{3/2}}{3/2} + C && \text{Integrate, using Eq. (1)} \\
 &&& \text{with } n = 1/2. \\
 &= \frac{1}{6} u^{3/2} + C && \text{Simpler form} \\
 &= \frac{1}{6} (4t-1)^{3/2} + C && \text{Replace } u \text{ by } 4t-1. \quad \blacksquare
 \end{aligned}$$

Substitution: Running the Chain Rule Backwards

The substitutions in Examples 1 and 2 are instances of the following general rule.

THEOREM 5 The Substitution Rule

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du.$$

Proof The rule is true because, by the Chain Rule, $F(g(x))$ is an antiderivative of $f(g(x)) \cdot g'(x)$ whenever F is an antiderivative of f :

$$\begin{aligned}
 \frac{d}{dx} F(g(x)) &= F'(g(x)) \cdot g'(x) && \text{Chain Rule} \\
 &= f(g(x)) \cdot g'(x). && \text{Because } F' = f
 \end{aligned}$$

If we make the substitution $u = g(x)$ then

$$\begin{aligned}
 \int f(g(x))g'(x) \, dx &= \int \frac{d}{dx} F(g(x)) \, dx \\
 &= F(g(x)) + C && \text{Fundamental Theorem} \\
 &= F(u) + C && u = g(x) \\
 &= \int F'(u) \, du && \text{Fundamental Theorem} \\
 &= \int f(u) \, du && F' = f \quad \blacksquare
 \end{aligned}$$

The Substitution Rule provides the following method to evaluate the integral

$$\int f(g(x))g'(x) dx,$$

when f and g' are continuous functions:

1. Substitute $u = g(x)$ and $du = g'(x) dx$ to obtain the integral

$$\int f(u) du.$$

2. Integrate with respect to u .
3. Replace u by $g(x)$ in the result.

EXAMPLE 3 Using Substitution

$$\begin{aligned}\int \cos(7\theta + 5) d\theta &= \int \cos u \cdot \frac{1}{7} du \\ &= \frac{1}{7} \int \cos u du \\ &= \frac{1}{7} \sin u + C \\ &= \frac{1}{7} \sin(7\theta + 5) + C\end{aligned}$$

Let $u = 7\theta + 5$, $du = 7 d\theta$,
($1/7$) $du = d\theta$.

With the $(1/7)$ out front, the
integral is now in standard form.

Integrate with respect to u ,
Table 4.2.

Replace u by $7\theta + 5$.

We can verify this solution by differentiating and checking that we obtain the original function $\cos(7\theta + 5)$. ■

EXAMPLE 4 Using Substitution

$$\begin{aligned}\int x^2 \sin(x^3) dx &= \int \sin(x^3) \cdot x^2 dx \\ &= \int \sin u \cdot \frac{1}{3} du \\ &= \frac{1}{3} \int \sin u du \\ &= \frac{1}{3} (-\cos u) + C \\ &= -\frac{1}{3} \cos(x^3) + C\end{aligned}$$

Let $u = x^3$,
 $du = 3x^2 dx$,
($1/3$) $du = x^2 dx$.

Integrate with respect to u .

Replace u by x^3 . ■

EXAMPLE 5 Using Identities and Substitution

$$\begin{aligned}
\int \frac{1}{\cos^2 2x} dx &= \int \sec^2 2x dx && \frac{1}{\cos 2x} = \sec 2x \\
&= \int \sec^2 u \cdot \frac{1}{2} du && u = 2x, \\
&= \frac{1}{2} \int \sec^2 u du && du = 2 dx, \\
&= \frac{1}{2} \tan u + C && dx = (1/2) du \\
&= \frac{1}{2} \tan 2x + C && \frac{d}{du} \tan u = \sec^2 u \\
&&& u = 2x
\end{aligned}$$

The success of the substitution method depends on finding a substitution that changes an integral we cannot evaluate directly into one that we can. If the first substitution fails, try to simplify the integrand further with an additional substitution or two (see Exercises 49 and 50). Alternatively, we can start fresh. There can be more than one good way to start, as in the next example.

EXAMPLE 6 Using Different Substitutions

Evaluate

$$\int \frac{2z dz}{\sqrt[3]{z^2 + 1}}.$$

Solution We can use the substitution method of integration as an exploratory tool: Substitute for the most troublesome part of the integrand and see how things work out. For the integral here, we might try $u = z^2 + 1$ or we might even press our luck and take u to be the entire cube root. Here is what happens in each case.

Solution 1: Substitute $u = z^2 + 1$.

$$\begin{aligned}
\int \frac{2z dz}{\sqrt[3]{z^2 + 1}} &= \int \frac{du}{u^{1/3}} && \text{Let } u = z^2 + 1, \\
&= \int u^{-1/3} du && du = 2z dz. \\
&= \frac{u^{2/3}}{2/3} + C && \text{In the form } \int u^n du \\
&= \frac{3}{2} u^{2/3} + C && \text{Integrate with respect to } u. \\
&= \frac{3}{2} (z^2 + 1)^{2/3} + C && \text{Replace } u \text{ by } z^2 + 1.
\end{aligned}$$

Solution 2: Substitute $u = \sqrt[3]{z^2 + 1}$ instead.

$$\begin{aligned}\int \frac{2z \, dz}{\sqrt[3]{z^2 + 1}} &= \int \frac{3u^2 \, du}{u} \\ &= 3 \int u \, du \\ &= 3 \cdot \frac{u^2}{2} + C \\ &= \frac{3}{2} (z^2 + 1)^{2/3} + C\end{aligned}$$

Let $u = \sqrt[3]{z^2 + 1}$,
 $u^3 = z^2 + 1$,
 $3u^2 \, du = 2z \, dz$.

Integrate with respect to u .

Replace u by $(z^2 + 1)^{1/3}$. ■

The Integrals of $\sin^2 x$ and $\cos^2 x$

Sometimes we can use trigonometric identities to transform integrals we do not know how to evaluate into ones we can use the substitution rule. Here is an example giving the integral formulas for $\sin^2 x$ and $\cos^2 x$ which arise frequently in applications.

EXAMPLE 7

$$\begin{aligned}\text{(a)} \quad \int \sin^2 x \, dx &= \int \frac{1 - \cos 2x}{2} \, dx && \sin^2 x = \frac{1 - \cos 2x}{2} \\ &= \frac{1}{2} \int (1 - \cos 2x) \, dx = \frac{1}{2} \int dx - \frac{1}{2} \int \cos 2x \, dx \\ &= \frac{1}{2} x - \frac{1}{2} \frac{\sin 2x}{2} + C = \frac{x}{2} - \frac{\sin 2x}{4} + C \\ \text{(b)} \quad \int \cos^2 x \, dx &= \int \frac{1 + \cos 2x}{2} \, dx && \cos^2 x = \frac{1 + \cos 2x}{2} \\ &= \frac{x}{2} + \frac{\sin 2x}{4} + C && \text{As in part (a), but with a sign change}\end{aligned}$$

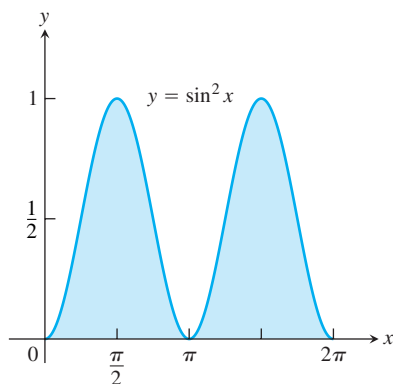


FIGURE 5.24 The area beneath the curve $y = \sin^2 x$ over $[0, 2\pi]$ equals π square units (Example 8).

EXAMPLE 8 Area Beneath the Curve $y = \sin^2 x$

Figure 5.24 shows the graph of $g(x) = \sin^2 x$ over the interval $[0, 2\pi]$. Find

- the definite integral of $g(x)$ over $[0, 2\pi]$.
- the area between the graph of the function and the x -axis over $[0, 2\pi]$.

Solution

- (a) From Example 7(a), the definite integral is

$$\begin{aligned}\int_0^{2\pi} \sin^2 x \, dx &= \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_0^{2\pi} = \left[\frac{2\pi}{2} - \frac{\sin 4\pi}{4} \right] - \left[\frac{0}{2} - \frac{\sin 0}{4} \right] \\ &= [\pi - 0] - [0 - 0] = \pi.\end{aligned}$$

- (b) The function $\sin^2 x$ is nonnegative, so the area is equal to the definite integral, or π . ■

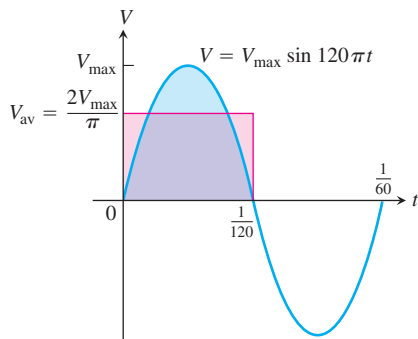


FIGURE 5.25 The graph of the voltage $V = V_{\max} \sin 120\pi t$ over a full cycle. Its average value over a half-cycle is $2V_{\max}/\pi$. Its average value over a full cycle is zero (Example 9).

EXAMPLE 9 Household Electricity

We can model the voltage in our home wiring with the sine function

$$V = V_{\max} \sin 120\pi t,$$

which expresses the voltage V in volts as a function of time t in seconds. The function runs through 60 cycles each second (its frequency is 60 hertz, or 60 Hz). The positive constant $V_{\max}$ (“vee max”) is the **peak voltage**.

The average value of V over the half-cycle from 0 to $1/120$ sec (see Figure 5.25) is

$$\begin{aligned} V_{\text{av}} &= \frac{1}{(1/120) - 0} \int_0^{1/120} V_{\max} \sin 120\pi t \, dt \\ &= 120V_{\max} \left[-\frac{1}{120\pi} \cos 120\pi t \right]_0^{1/120} \\ &= \frac{V_{\max}}{\pi} [-\cos \pi + \cos 0] \\ &= \frac{2V_{\max}}{\pi}. \end{aligned}$$

The average value of the voltage over a full cycle is zero, as we can see from Figure 5.25. (Also see Exercise 63.) If we measured the voltage with a standard moving-coil galvanometer, the meter would read zero.

To measure the voltage effectively, we use an instrument that measures the square root of the average value of the square of the voltage, namely

$$V_{\text{rms}} = \sqrt{(V^2)_{\text{av}}}.$$

The subscript “rms” (read the letters separately) stands for “root mean square.” Since the average value of $V^2 = (V_{\max})^2 \sin^2 120\pi t$ over a cycle is

$$(V^2)_{\text{av}} = \frac{1}{(1/60) - 0} \int_0^{1/60} (V_{\max})^2 \sin^2 120\pi t \, dt = \frac{(V_{\max})^2}{2},$$

(Exercise 63, part c), the rms voltage is

$$V_{\text{rms}} = \sqrt{\frac{(V_{\max})^2}{2}} = \frac{V_{\max}}{\sqrt{2}}.$$

The values given for household currents and voltages are always rms values. Thus, “115 volts ac” means that the rms voltage is 115. The peak voltage, obtained from the last equation, is

$$V_{\max} = \sqrt{2} V_{\text{rms}} = \sqrt{2} \cdot 115 \approx 163 \text{ volts},$$

which is considerably higher. ■

EXERCISES 5.5

Evaluating Integrals

Evaluate the indefinite integrals in Exercises 1–12 by using the given substitutions to reduce the integrals to standard form.

1. $\int \sin 3x \, dx, \quad u = 3x$

2. $\int x \sin(2x^2) \, dx, \quad u = 2x^2$

3. $\int \sec 2t \tan 2t \, dt, \quad u = 2t$

4. $\int \left(1 - \cos \frac{t}{2}\right)^2 \sin \frac{t}{2} \, dt, \quad u = 1 - \cos \frac{t}{2}$

5. $\int 28(7x - 2)^{-5} dx, \quad u = 7x - 2$
6. $\int x^3(x^4 - 1)^2 dx, \quad u = x^4 - 1$
7. $\int \frac{9r^2 dr}{\sqrt{1 - r^3}}, \quad u = 1 - r^3$
8. $\int 12(y^4 + 4y^2 + 1)^2(y^3 + 2y) dy, \quad u = y^4 + 4y^2 + 1$
9. $\int \sqrt{x} \sin^2(x^{3/2} - 1) dx, \quad u = x^{3/2} - 1$
10. $\int \frac{1}{x^2} \cos^2\left(\frac{1}{x}\right) dx, \quad u = -\frac{1}{x}$
11. $\int \csc^2 2\theta \cot 2\theta d\theta$
 a. Using $u = \cot 2\theta$ b. Using $u = \csc 2\theta$
12. $\int \frac{dx}{\sqrt{5x + 8}}$
 a. Using $u = 5x + 8$ b. Using $u = \sqrt{5x + 8}$

Evaluate the integrals in Exercises 13–48.

13. $\int \sqrt{3 - 2s} ds$ 14. $\int (2x + 1)^3 dx$
15. $\int \frac{1}{\sqrt{5s + 4}} ds$ 16. $\int \frac{3 dx}{(2 - x)^2}$
17. $\int \theta \sqrt[4]{1 - \theta^2} d\theta$ 18. $\int 8\theta \sqrt[3]{\theta^2 - 1} d\theta$
19. $\int 3y \sqrt{7 - 3y^2} dy$ 20. $\int \frac{4y dy}{\sqrt{2y^2 + 1}}$
21. $\int \frac{1}{\sqrt{x}(1 + \sqrt{x})^2} dx$ 22. $\int \frac{(1 + \sqrt{x})^3}{\sqrt{x}} dx$
23. $\int \cos(3z + 4) dz$ 24. $\int \sin(8z - 5) dz$
25. $\int \sec^2(3x + 2) dx$ 26. $\int \tan^2 x \sec^2 x dx$
27. $\int \sin^5 \frac{x}{3} \cos \frac{x}{3} dx$ 28. $\int \tan^7 \frac{x}{2} \sec^2 \frac{x}{2} dx$
29. $\int r^2 \left(\frac{r^3}{18} - 1\right)^5 dr$ 30. $\int r^4 \left(7 - \frac{r^5}{10}\right)^3 dr$
31. $\int x^{1/2} \sin(x^{3/2} + 1) dx$ 32. $\int x^{1/3} \sin(x^{4/3} - 8) dx$
33. $\int \sec\left(v + \frac{\pi}{2}\right) \tan\left(v + \frac{\pi}{2}\right) dv$
34. $\int \csc\left(\frac{v - \pi}{2}\right) \cot\left(\frac{v - \pi}{2}\right) dv$
35. $\int \frac{\sin(2t + 1)}{\cos^2(2t + 1)} dt$ 36. $\int \frac{6 \cos t}{(2 + \sin t)^3} dt$
37. $\int \sqrt{\cot y} \csc^2 y dy$ 38. $\int \frac{\sec z \tan z}{\sqrt{\sec z}} dz$
39. $\int \frac{1}{t^2} \cos\left(\frac{1}{t} - 1\right) dt$ 40. $\int \frac{1}{\sqrt{t}} \cos(\sqrt{t} + 3) dt$
41. $\int \frac{1}{\theta^2} \sin \frac{1}{\theta} \cos \frac{1}{\theta} d\theta$ 42. $\int \frac{\cos \sqrt{\theta}}{\sqrt{\theta} \sin^2 \sqrt{\theta}} d\theta$
43. $\int (s^3 + 2s^2 - 5s + 5)(3s^2 + 4s - 5) ds$
44. $\int (\theta^4 - 2\theta^2 + 8\theta - 2)(\theta^3 - \theta + 2) d\theta$
45. $\int t^3(1 + t^4)^3 dt$ 46. $\int \sqrt{\frac{x - 1}{x^5}} dx$
47. $\int x^3 \sqrt{x^2 + 1} dx$ 48. $\int 3x^5 \sqrt{x^3 + 1} dx$

Simplifying Integrals Step by Step

If you do not know what substitution to make, try reducing the integral step by step, using a trial substitution to simplify the integral a bit and then another to simplify it some more. You will see what we mean if you try the sequences of substitutions in Exercises 49 and 50.

49. $\int \frac{18 \tan^2 x \sec^2 x}{(2 + \tan^3 x)^2} dx$
 a. $u = \tan x$, followed by $v = u^3$, then by $w = 2 + v$
 b. $u = \tan^3 x$, followed by $v = 2 + u$
 c. $u = 2 + \tan^3 x$
50. $\int \sqrt{1 + \sin^2(x - 1)} \sin(x - 1) \cos(x - 1) dx$
 a. $u = x - 1$, followed by $v = \sin u$, then by $w = 1 + v^2$
 b. $u = \sin(x - 1)$, followed by $v = 1 + u^2$
 c. $u = 1 + \sin^2(x - 1)$

Evaluate the integrals in Exercises 51 and 52.

51. $\int \frac{(2r - 1) \cos \sqrt{3(2r - 1)^2 + 6}}{\sqrt{3(2r - 1)^2 + 6}} dr$
52. $\int \frac{\sin \sqrt{\theta}}{\sqrt{\theta} \cos^3 \sqrt{\theta}} d\theta$

Initial Value Problems

Solve the initial value problems in Exercises 53–58.

53. $\frac{ds}{dt} = 12t(3t^2 - 1)^3, \quad s(1) = 3$
54. $\frac{dy}{dx} = 4x(x^2 + 8)^{-1/3}, \quad y(0) = 0$
55. $\frac{ds}{dt} = 8 \sin^2\left(t + \frac{\pi}{12}\right), \quad s(0) = 8$
56. $\frac{dr}{d\theta} = 3 \cos^2\left(\frac{\pi}{4} - \theta\right), \quad r(0) = \frac{\pi}{8}$

57. $\frac{d^2s}{dt^2} = -4 \sin\left(2t - \frac{\pi}{2}\right), \quad s'(0) = 100, \quad s(0) = 0$

58. $\frac{d^2y}{dx^2} = 4 \sec^2 2x \tan 2x, \quad y'(0) = 4, \quad y(0) = -1$

59. The velocity of a particle moving back and forth on a line is $v = ds/dt = 6 \sin 2t$ m/sec for all t . If $s = 0$ when $t = 0$, find the value of s when $t = \pi/2$ sec.

60. The acceleration of a particle moving back and forth on a line is $a = d^2s/dt^2 = \pi^2 \cos \pi t$ m/sec² for all t . If $s = 0$ and $v = 8$ m/sec when $t = 0$, find s when $t = 1$ sec.

Theory and Examples

61. It looks as if we can integrate $2 \sin x \cos x$ with respect to x in three different ways:

a. $\int 2 \sin x \cos x \, dx = \int 2u \, du \quad u = \sin x,$
 $= u^2 + C_1 = \sin^2 x + C_1$

b. $\int 2 \sin x \cos x \, dx = \int -2u \, du \quad u = \cos x,$
 $= -u^2 + C_2 = -\cos^2 x + C_2$

c. $\int 2 \sin x \cos x \, dx = \int \sin 2x \, dx \quad 2 \sin x \cos x = \sin 2x$
 $= -\frac{\cos 2x}{2} + C_3.$

Can all three integrations be correct? Give reasons for your answer.

62. The substitution $u = \tan x$ gives

$$\int \sec^2 x \tan x \, dx = \int u \, du = \frac{u^2}{2} + C = \frac{\tan^2 x}{2} + C.$$

The substitution $u = \sec x$ gives

$$\int \sec^2 x \tan x \, dx = \int u \, du = \frac{u^2}{2} + C = \frac{\sec^2 x}{2} + C.$$

Can both integrations be correct? Give reasons for your answer.

63. (Continuation of Example 9.)

a. Show by evaluating the integral in the expression

$$\frac{1}{(1/60) - 0} \int_0^{1/60} V_{\max} \sin 120 \pi t \, dt$$

that the average value of $V = V_{\max} \sin 120 \pi t$ over a full cycle is zero.

b. The circuit that runs your electric stove is rated 240 volts rms. What is the peak value of the allowable voltage?

c. Show that

$$\int_0^{1/60} (V_{\max})^2 \sin^2 120 \pi t \, dt = \frac{(V_{\max})^2}{120}.$$

5.6

Substitution and Area Between Curves

There are two methods for evaluating a definite integral by substitution. The first method is to find an antiderivative using substitution, and then to evaluate the definite integral by applying the Fundamental Theorem. We used this method in Examples 8 and 9 of the preceding section. The second method extends the process of substitution directly to *definite* integrals. We apply the new formula introduced here to the problem of computing the area between two curves.

Substitution Formula

In the following formula, the limits of integration change when the variable of integration is changed by substitution.

THEOREM 6 Substitution in Definite Integrals

If g' is continuous on the interval $[a, b]$ and f is continuous on the range of g , then

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du$$

Proof Let F denote any antiderivative of f . Then,

$$\begin{aligned}
 \int_a^b f(g(x)) \cdot g'(x) \, dx &= F(g(x)) \Big|_{x=a}^{x=b} && \frac{d}{dx} F(g(x)) \\
 &= F(g(b)) - F(g(a)) && = F'(g(x))g'(x) \\
 &= F(u) \Big|_{u=g(a)}^{u=g(b)} && = f(g(x))g'(x) \\
 &= \int_{g(a)}^{g(b)} f(u) \, du. && \text{Fundamental Theorem, Part 2} \quad \blacksquare
 \end{aligned}$$

To use the formula, make the same u -substitution $u = g(x)$ and $du = g'(x) \, dx$ you would use to evaluate the corresponding indefinite integral. Then integrate the transformed integral with respect to u from the value $g(a)$ (the value of u at $x = a$) to the value $g(b)$ (the value of u at $x = b$).

EXAMPLE 1 Substitution by Two Methods

Evaluate $\int_{-1}^1 3x^2 \sqrt{x^3 + 1} \, dx$.

Solution We have two choices.

Method 1: Transform the integral and evaluate the transformed integral with the transformed limits given in Theorem 6.

$$\begin{aligned}
 \int_{-1}^1 3x^2 \sqrt{x^3 + 1} \, dx &= \int_0^2 \sqrt{u} \, du && \begin{array}{l} \text{Let } u = x^3 + 1, \, du = 3x^2 \, dx. \\ \text{When } x = -1, \, u = (-1)^3 + 1 = 0. \\ \text{When } x = 1, \, u = (1)^3 + 1 = 2. \end{array} \\
 &= \frac{2}{3} u^{3/2} \Big|_0^2 && \text{Evaluate the new definite integral.} \\
 &= \frac{2}{3} \left[2^{3/2} - 0^{3/2} \right] = \frac{2}{3} \left[2\sqrt{2} \right] = \frac{4\sqrt{2}}{3}
 \end{aligned}$$

Method 2: Transform the integral as an indefinite integral, integrate, change back to x , and use the original x -limits.

$$\begin{aligned}
 \int 3x^2 \sqrt{x^3 + 1} \, dx &= \int \sqrt{u} \, du && \text{Let } u = x^3 + 1, \, du = 3x^2 \, dx. \\
 &= \frac{2}{3} u^{3/2} + C && \text{Integrate with respect to } u. \\
 &= \frac{2}{3} (x^3 + 1)^{3/2} + C && \text{Replace } u \text{ by } x^3 + 1. \\
 \int_{-1}^1 3x^2 \sqrt{x^3 + 1} \, dx &= \frac{2}{3} (x^3 + 1)^{3/2} \Big|_{-1}^1 && \text{Use the integral just found,} \\
 &= \frac{2}{3} \left[((1)^3 + 1)^{3/2} - ((-1)^3 + 1)^{3/2} \right] && \text{with limits of integration for } x. \\
 &= \frac{2}{3} \left[2^{3/2} - 0^{3/2} \right] = \frac{2}{3} \left[2\sqrt{2} \right] = \frac{4\sqrt{2}}{3} \quad \blacksquare
 \end{aligned}$$

Which method is better—evaluating the transformed definite integral with transformed limits using Theorem 6, or transforming the integral, integrating, and transforming back to use the original limits of integration? In Example 1, the first method seems easier, but that is not always the case. Generally, it is best to know both methods and to use whichever one seems better at the time.

EXAMPLE 2 Using the Substitution Formula

$$\begin{aligned}
 \int_{\pi/4}^{\pi/2} \cot \theta \csc^2 \theta \, d\theta &= \int_1^0 u \cdot (-du) && \text{Let } u = \cot \theta, \, du = -\csc^2 \theta \, d\theta, \\
 &= -\int_1^0 u \, du && \quad -du = \csc^2 \theta \, d\theta. \\
 &= -\left[\frac{u^2}{2} \right]_1^0 && \text{When } \theta = \pi/4, u = \cot(\pi/4) = 1. \\
 &= -\left[\frac{(0)^2}{2} - \frac{(1)^2}{2} \right] = \frac{1}{2} && \text{When } \theta = \pi/2, u = \cot(\pi/2) = 0.
 \end{aligned}$$

Definite Integrals of Symmetric Functions

The Substitution Formula in Theorem 6 simplifies the calculation of definite integrals of even and odd functions (Section 1.4) over a symmetric interval $[-a, a]$ (Figure 5.26).

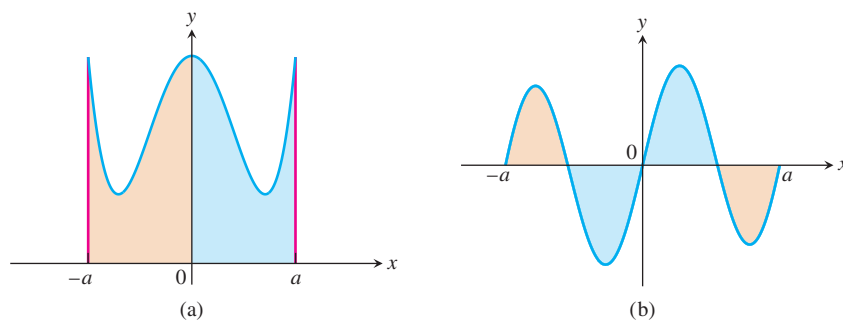


FIGURE 5.26 (a) f even, $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$ (b) f odd, $\int_{-a}^a f(x) \, dx = 0$

Theorem 7

Let f be continuous on the symmetric interval $[-a, a]$.

(a) If f is even, then $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$.

(b) If f is odd, then $\int_{-a}^a f(x) \, dx = 0$.

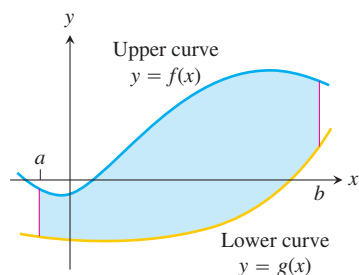


FIGURE 5.27 The region between the curves $y = f(x)$ and $y = g(x)$ and the lines $x = a$ and $x = b$.

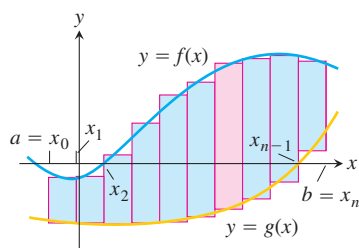


FIGURE 5.28 We approximate the region with rectangles perpendicular to the x -axis.

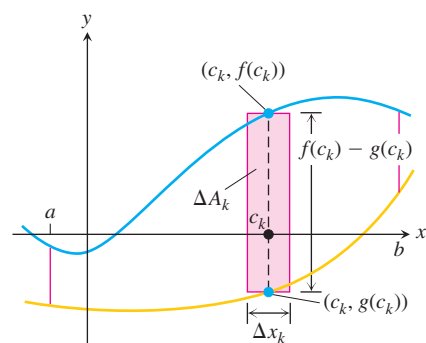


FIGURE 5.29 The area ΔA_k of the k th rectangle is the product of its height, $f(c_k) - g(c_k)$, and its width, Δx_k .

Proof of Part (a)

$$\begin{aligned}
 \int_{-a}^a f(x) \, dx &= \int_{-a}^0 f(x) \, dx + \int_0^a f(x) \, dx && \text{Additivity Rule for Definite Integrals} \\
 &= -\int_0^{-a} f(x) \, dx + \int_0^a f(x) \, dx && \text{Order of Integration Rule} \\
 &= -\int_0^a f(-u)(-du) + \int_0^a f(x) \, dx && \text{Let } u = -x, du = -dx. \\
 &&& \text{When } x = 0, u = 0. \\
 &&& \text{When } x = -a, u = a. \\
 &= \int_0^a f(-u) \, du + \int_0^a f(x) \, dx \\
 &= \int_0^a f(u) \, du + \int_0^a f(x) \, dx && f \text{ is even, so } f(-u) = f(u). \\
 &= 2 \int_0^a f(x) \, dx
 \end{aligned}$$

The proof of part (b) is entirely similar and you are asked to give it in Exercise 86. ■

The assertions of Theorem 7 remain true when f is an integrable function (rather than having the stronger property of being continuous), but the proof is somewhat more difficult and best left to a more advanced course.

EXAMPLE 3 Integral of an Even Function

Evaluate $\int_{-2}^2 (x^4 - 4x^2 + 6) \, dx$.

Solution Since $f(x) = x^4 - 4x^2 + 6$ satisfies $f(-x) = f(x)$, it is even on the symmetric interval $[-2, 2]$, so

$$\begin{aligned}
 \int_{-2}^2 (x^4 - 4x^2 + 6) \, dx &= 2 \int_0^2 (x^4 - 4x^2 + 6) \, dx \\
 &= 2 \left[\frac{x^5}{5} - \frac{4}{3}x^3 + 6x \right]_0^2 \\
 &= 2 \left(\frac{32}{5} - \frac{32}{3} + 12 \right) = \frac{232}{15}.
 \end{aligned}$$

Areas Between Curves

Suppose we want to find the area of a region that is bounded above by the curve $y = f(x)$, below by the curve $y = g(x)$, and on the left and right by the lines $x = a$ and $x = b$ (Figure 5.27). The region might accidentally have a shape whose area we could find with geometry, but if f and g are arbitrary continuous functions, we usually have to find the area with an integral.

To see what the integral should be, we first approximate the region with n vertical rectangles based on a partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ (Figure 5.28). The area of the k th rectangle (Figure 5.29) is

$$\Delta A_k = \text{height} \times \text{width} = [f(c_k) - g(c_k)] \Delta x_k.$$

We then approximate the area of the region by adding the areas of the n rectangles:

$$A \approx \sum_{k=1}^n \Delta A_k = \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k. \quad \text{Riemann Sum}$$

As $\|P\| \rightarrow 0$, the sums on the right approach the limit $\int_a^b [f(x) - g(x)] dx$ because f and g are continuous. We take the area of the region to be the value of this integral. That is,

$$A = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k = \int_a^b [f(x) - g(x)] dx.$$

DEFINITION Area Between Curves

If f and g are continuous with $f(x) \geq g(x)$ throughout $[a, b]$, then the **area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b** is the integral of $(f - g)$ from a to b :

$$A = \int_a^b [f(x) - g(x)] dx.$$

When applying this definition it is helpful to graph the curves. The graph reveals which curve is the upper curve f and which is the lower curve g . It also helps you find the limits of integration if they are not already known. You may need to find where the curves intersect to determine the limits of integration, and this may involve solving the equation $f(x) = g(x)$ for values of x . Then you can integrate the function $f - g$ for the area between the intersections.

EXAMPLE 4 Area Between Intersecting Curves

Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

Solution First we sketch the two curves (Figure 5.30). The limits of integration are found by solving $y = 2 - x^2$ and $y = -x$ simultaneously for x .

$$\begin{aligned} 2 - x^2 &= -x && \text{Equate } f(x) \text{ and } g(x). \\ x^2 - x - 2 &= 0 && \text{Rewrite.} \\ (x + 1)(x - 2) &= 0 && \text{Factor.} \\ x = -1, \quad x = 2. &&& \text{Solve.} \end{aligned}$$

The region runs from $x = -1$ to $x = 2$. The limits of integration are $a = -1$, $b = 2$. The area between the curves is

$$\begin{aligned} A &= \int_a^b [f(x) - g(x)] dx = \int_{-1}^2 [(2 - x^2) - (-x)] dx \\ &= \int_{-1}^2 (2 + x - x^2) dx = \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2 \\ &= \left(4 + \frac{4}{2} - \frac{8}{3} \right) - \left(-2 + \frac{1}{2} + \frac{1}{3} \right) = \frac{9}{2} \end{aligned}$$

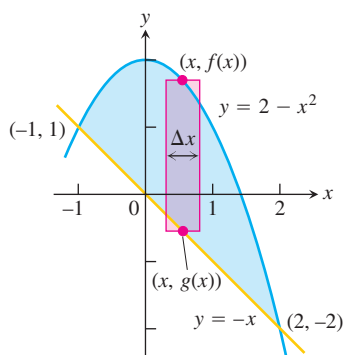


FIGURE 5.30 The region in Example 4 with a typical approximating rectangle.

HISTORICAL BIOGRAPHY

Richard Dedekind
(1831–1916)

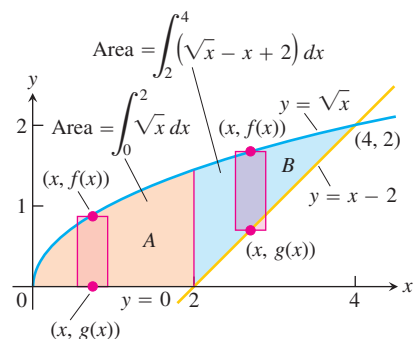


FIGURE 5.31 When the formula for a bounding curve changes, the area integral changes to become the sum of integrals to match, one integral for each of the shaded regions shown here for Example 5.

If the formula for a bounding curve changes at one or more points, we subdivide the region into subregions that correspond to the formula changes and apply the formula for the area between curves to each subregion.

EXAMPLE 5 Changing the Integral to Match a Boundary Change

Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$.

Solution The sketch (Figure 5.31) shows that the region's upper boundary is the graph of $f(x) = \sqrt{x}$. The lower boundary changes from $g(x) = 0$ for $0 \leq x \leq 2$ to $g(x) = x - 2$ for $2 \leq x \leq 4$ (there is agreement at $x = 2$). We subdivide the region at $x = 2$ into subregions A and B , shown in Figure 5.31.

The limits of integration for region A are $a = 0$ and $b = 2$. The left-hand limit for region B is $a = 2$. To find the right-hand limit, we solve the equations $y = \sqrt{x}$ and $y = x - 2$ simultaneously for x :

$$\begin{aligned} \sqrt{x} &= x - 2 && \text{Equate } f(x) \text{ and } g(x). \\ x &= (x - 2)^2 = x^2 - 4x + 4 && \text{Square both sides.} \\ x^2 - 5x + 4 &= 0 && \text{Rewrite.} \\ (x - 1)(x - 4) &= 0 && \text{Factor.} \\ x &= 1, \quad x = 4. && \text{Solve.} \end{aligned}$$

Only the value $x = 4$ satisfies the equation $\sqrt{x} = x - 2$. The value $x = 1$ is an extraneous root introduced by squaring. The right-hand limit is $b = 4$.

$$\begin{aligned} \text{For } 0 \leq x \leq 2: \quad f(x) - g(x) &= \sqrt{x} - 0 = \sqrt{x} \\ \text{For } 2 \leq x \leq 4: \quad f(x) - g(x) &= \sqrt{x} - (x - 2) = \sqrt{x} - x + 2 \end{aligned}$$

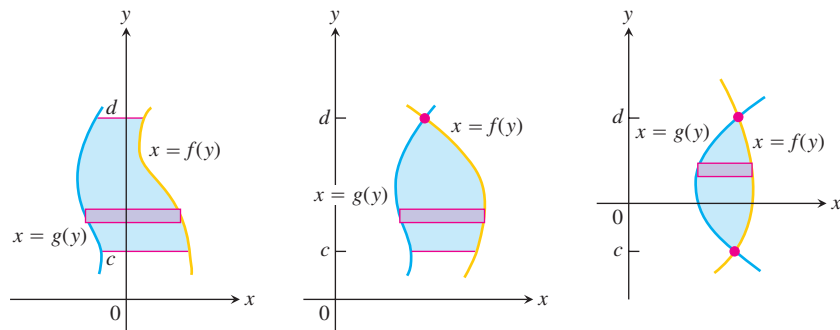
We add the area of subregions A and B to find the total area:

$$\begin{aligned} \text{Total area} &= \underbrace{\int_0^2 \sqrt{x} \, dx}_{\text{area of } A} + \underbrace{\int_2^4 (\sqrt{x} - x + 2) \, dx}_{\text{area of } B} \\ &= \left[\frac{2}{3} x^{3/2} \right]_0^2 + \left[\frac{2}{3} x^{3/2} - \frac{x^2}{2} + 2x \right]_2^4 \\ &= \frac{2}{3} (2)^{3/2} - 0 + \left(\frac{2}{3} (4)^{3/2} - 8 + 8 \right) - \left(\frac{2}{3} (2)^{3/2} - 2 + 4 \right) \\ &= \frac{2}{3} (8) - 2 = \frac{10}{3}. \end{aligned}$$

Integration with Respect to y

If a region's bounding curves are described by functions of y , the approximating rectangles are horizontal instead of vertical and the basic formula has y in place of x .

For regions like these



use the formula

$$A = \int_c^d [f(y) - g(y)] dy.$$

In this equation f always denotes the right-hand curve and g the left-hand curve, so $f(y) - g(y)$ is nonnegative.

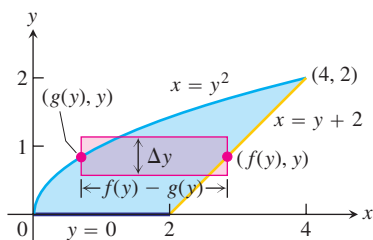


FIGURE 5.32 It takes two integrations to find the area of this region if we integrate with respect to x . It takes only one if we integrate with respect to y (Example 6).

EXAMPLE 6 Find the area of the region in Example 5 by integrating with respect to y .

Solution We first sketch the region and a typical *horizontal* rectangle based on a partition of an interval of y -values (Figure 5.32). The region's right-hand boundary is the line $x = y + 2$, so $f(y) = y + 2$. The left-hand boundary is the curve $x = y^2$, so $g(y) = y^2$. The lower limit of integration is $y = 0$. We find the upper limit by solving $x = y + 2$ and $x = y^2$ simultaneously for y :

$$\begin{aligned} y + 2 &= y^2 && \text{Equate } f(y) = y + 2 \\ &&& \text{and } g(y) = y^2. \\ y^2 - y - 2 &= 0 && \text{Rewrite.} \\ (y + 1)(y - 2) &= 0 && \text{Factor.} \\ y = -1, \quad y = 2 &&& \text{Solve.} \end{aligned}$$

The upper limit of integration is $b = 2$. (The value $y = -1$ gives a point of intersection *below* the x -axis.)

The area of the region is

$$\begin{aligned} A &= \int_a^b [f(y) - g(y)] dy = \int_0^2 [y + 2 - y^2] dy \\ &= \int_0^2 [2 + y - y^2] dy \\ &= \left[2y + \frac{y^2}{2} - \frac{y^3}{3} \right]_0^2 \\ &= 4 + \frac{4}{2} - \frac{8}{3} = \frac{10}{3}. \end{aligned}$$

This is the result of Example 5, found with less work. ■

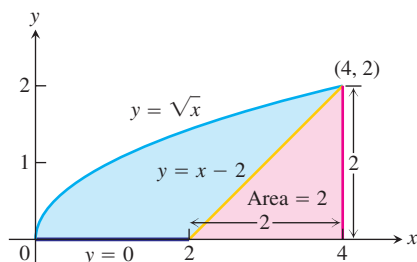


FIGURE 5.33 The area of the blue region is the area under the parabola $y = \sqrt{x}$ minus the area of the triangle (Example 7).

Combining Integrals with Formulas from Geometry

The fastest way to find an area may be to combine calculus and geometry.

EXAMPLE 7 The Area of the Region in Example 5 Found the Fastest Way

Find the area of the region in Example 5.

Solution The area we want is the area between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$, and the x -axis, *minus* the area of a triangle with base 2 and height 2 (Figure 5.33):

$$\begin{aligned} \text{Area} &= \int_0^4 \sqrt{x} \, dx - \frac{1}{2}(2)(2) \\ &= \frac{2}{3}x^{3/2} \Big|_0^4 - 2 \\ &= \frac{2}{3}(8) - 0 - 2 = \frac{10}{3}. \end{aligned}$$

Conclusion from Examples 5–7 It is sometimes easier to find the area between two curves by integrating with respect to y instead of x . Also, it may help to combine geometry and calculus. After sketching the region, take a moment to think about the best way to proceed.

EXERCISES 5.6

Evaluating Definite Integrals

Use the Substitution Formula in Theorem 6 to evaluate the integrals in Exercises 1–24.

1. a. $\int_0^3 \sqrt{y+1} \, dy$

b. $\int_{-1}^0 \sqrt{y+1} \, dy$

2. a. $\int_0^1 r\sqrt{1-r^2} \, dr$

b. $\int_{-1}^1 r\sqrt{1-r^2} \, dr$

3. a. $\int_0^{\pi/4} \tan x \sec^2 x \, dx$

b. $\int_{-\pi/4}^0 \tan x \sec^2 x \, dx$

4. a. $\int_0^{\pi} 3 \cos^2 x \sin x \, dx$

b. $\int_{2\pi}^{3\pi} 3 \cos^2 x \sin x \, dx$

5. a. $\int_0^1 t^3(1+t^4)^3 \, dt$

b. $\int_{-1}^1 t^3(1+t^4)^3 \, dt$

6. a. $\int_0^{\sqrt{7}} t(t^2+1)^{1/3} \, dt$

b. $\int_{-\sqrt{7}}^0 t(t^2+1)^{1/3} \, dt$

7. a. $\int_{-1}^1 \frac{5r}{(4+r^2)^2} \, dr$

b. $\int_0^1 \frac{5r}{(4+r^2)^2} \, dr$

8. a. $\int_0^1 \frac{10\sqrt{v}}{(1+v^{3/2})^2} \, dv$

b. $\int_1^4 \frac{10\sqrt{v}}{(1+v^{3/2})^2} \, dv$

9. a. $\int_0^{\sqrt{3}} \frac{4x}{\sqrt{x^2+1}} \, dx$

b. $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{4x}{\sqrt{x^2+1}} \, dx$

10. a. $\int_0^1 \frac{x^3}{\sqrt{x^4+9}} \, dx$

b. $\int_{-1}^0 \frac{x^3}{\sqrt{x^4+9}} \, dx$

11. a. $\int_0^{\pi/6} (1 - \cos 3t) \sin 3t \, dt$

b. $\int_{\pi/6}^{\pi/3} (1 - \cos 3t) \sin 3t \, dt$

12. a. $\int_{-\pi/2}^0 \left(2 + \tan \frac{t}{2}\right) \sec^2 \frac{t}{2} \, dt$

b. $\int_{-\pi/2}^{\pi/2} \left(2 + \tan \frac{t}{2}\right) \sec^2 \frac{t}{2} \, dt$

13. a. $\int_0^{2\pi} \frac{\cos z}{\sqrt{4+3\sin z}} \, dz$

b. $\int_{-\pi}^{\pi} \frac{\cos z}{\sqrt{4+3\sin z}} \, dz$

14. a. $\int_{-\pi/2}^0 \frac{\sin w}{(3+2\cos w)^2} \, dw$

b. $\int_0^{\pi/2} \frac{\sin w}{(3+2\cos w)^2} \, dw$

15. $\int_0^1 \sqrt{t^5+2t} (5t^4+2) \, dt$

16. $\int_1^4 \frac{dy}{2\sqrt{y}(1+\sqrt{y})^2}$

17. $\int_0^{\pi/6} \cos^{-3} 2\theta \sin 2\theta \, d\theta$

18. $\int_{\pi}^{3\pi/2} \cot^5 \left(\frac{\theta}{6}\right) \sec^2 \left(\frac{\theta}{6}\right) \, d\theta$

19. $\int_0^{\pi} 5(5-4\cos t)^{1/4} \sin t \, dt$

20. $\int_0^{\pi/4} (1-\sin 2t)^{3/2} \cos 2t \, dt$

21. $\int_0^1 (4y-y^2+4y^3+1)^{-2/3} (12y^2-2y+4) \, dy$

22. $\int_0^1 (y^3+6y^2-12y+9)^{-1/2} (y^2+4y-4) \, dy$

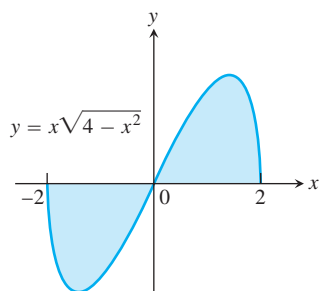
23. $\int_0^{\sqrt[3]{\pi^2}} \sqrt{\theta} \cos^2(\theta^{3/2}) d\theta$

24. $\int_{-1}^{-1/2} t^{-2} \sin^2\left(1 + \frac{1}{t}\right) dt$

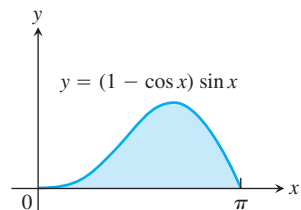
Area

Find the total areas of the shaded regions in Exercises 25–40.

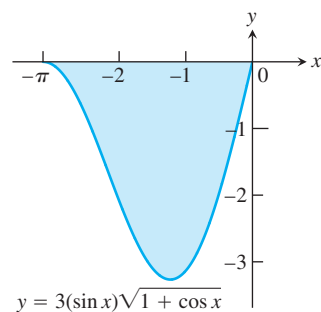
25.



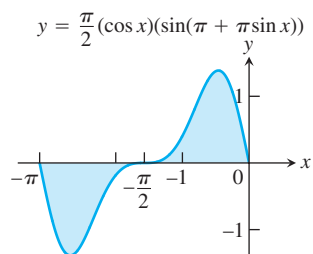
26.



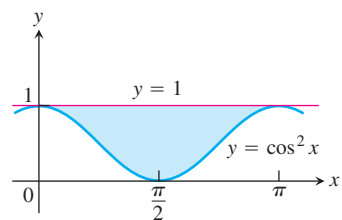
27.



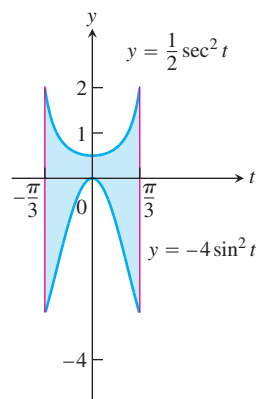
28.



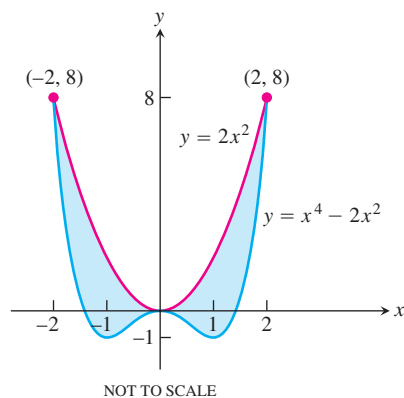
29.



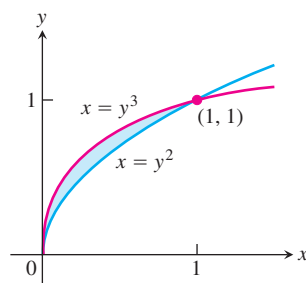
30.



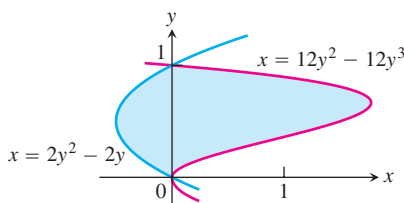
31.



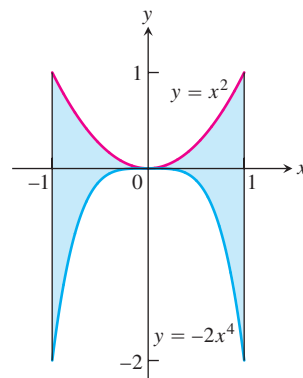
32.



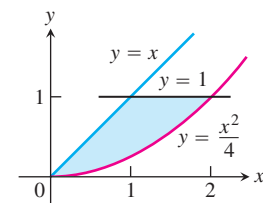
33.

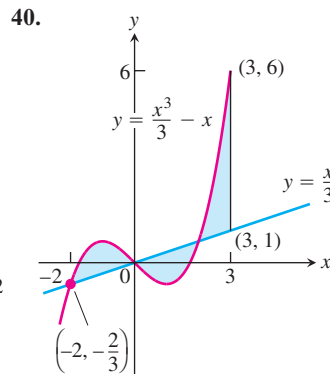
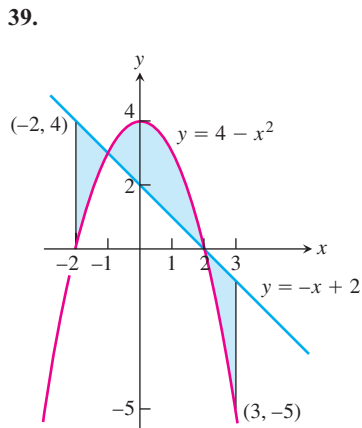
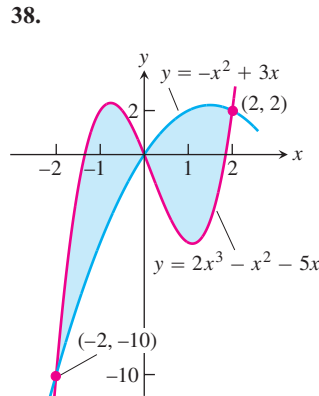
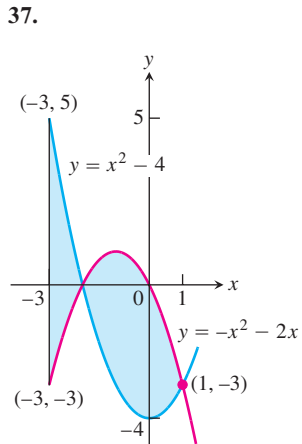
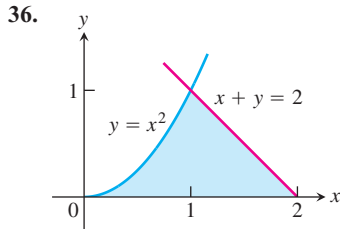


34.



35.





Find the areas of the regions enclosed by the lines and curves in Exercises 41–50.

41. $y = x^2 - 2$ and $y = 2$
42. $y = 2x - x^2$ and $y = -3$
43. $y = x^4$ and $y = 8x$
44. $y = x^2 - 2x$ and $y = x$
45. $y = x^2$ and $y = -x^2 + 4x$
46. $y = 7 - 2x^2$ and $y = x^2 + 4$
47. $y = x^4 - 4x^2 + 4$ and $y = x^2$
48. $y = x\sqrt{a^2 - x^2}$, $a > 0$, and $y = 0$

49. $y = \sqrt{|x|}$ and $5y = x + 6$ (How many intersection points are there?)

50. $y = |x^2 - 4|$ and $y = (x^2/2) + 4$

Find the areas of the regions enclosed by the lines and curves in Exercises 51–58.

51. $x = 2y^2$, $x = 0$, and $y = 3$
52. $x = y^2$ and $x = y + 2$
53. $y^2 - 4x = 4$ and $4x - y = 16$
54. $x - y^2 = 0$ and $x + 2y^2 = 3$
55. $x + y^2 = 0$ and $x + 3y^2 = 2$
56. $x - y^{2/3} = 0$ and $x + y^4 = 2$
57. $x = y^2 - 1$ and $x = |y|\sqrt{1 - y^2}$
58. $x = y^3 - y^2$ and $x = 2y$

Find the areas of the regions enclosed by the curves in Exercises 59–62.

59. $4x^2 + y = 4$ and $x^4 - y = 1$
60. $x^3 - y = 0$ and $3x^2 - y = 4$
61. $x + 4y^2 = 4$ and $x + y^4 = 1$, for $x \geq 0$
62. $x + y^2 = 3$ and $4x + y^2 = 0$

Find the areas of the regions enclosed by the lines and curves in Exercises 63–70.

63. $y = 2 \sin x$ and $y = \sin 2x$, $0 \leq x \leq \pi$
64. $y = 8 \cos x$ and $y = \sec^2 x$, $-\pi/3 \leq x \leq \pi/3$
65. $y = \cos(\pi x/2)$ and $y = 1 - x^2$
66. $y = \sin(\pi x/2)$ and $y = x$
67. $y = \sec^2 x$, $y = \tan^2 x$, $x = -\pi/4$, and $x = \pi/4$
68. $x = \tan^2 y$ and $x = -\tan^2 y$, $-\pi/4 \leq y \leq \pi/4$
69. $x = 3 \sin y \sqrt{\cos y}$ and $x = 0$, $0 \leq y \leq \pi/2$
70. $y = \sec^2(\pi x/3)$ and $y = x^{1/3}$, $-1 \leq x \leq 1$

71. Find the area of the propeller-shaped region enclosed by the curve $x - y^3 = 0$ and the line $x - y = 0$.

72. Find the area of the propeller-shaped region enclosed by the curves $x - y^{1/3} = 0$ and $x - y^{1/5} = 0$.

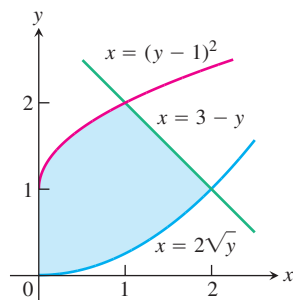
73. Find the area of the region in the first quadrant bounded by the line $y = x$, the line $x = 2$, the curve $y = 1/x^2$, and the x -axis.

74. Find the area of the “triangular” region in the first quadrant bounded on the left by the y -axis and on the right by the curves $y = \sin x$ and $y = \cos x$.

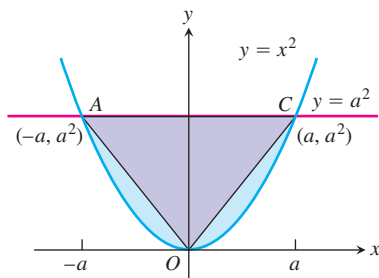
75. The region bounded below by the parabola $y = x^2$ and above by the line $y = 4$ is to be partitioned into two subsections of equal area by cutting across it with the horizontal line $y = c$.

- a. Sketch the region and draw a line $y = c$ across it that looks about right. In terms of c , what are the coordinates of the points where the line and parabola intersect? Add them to your figure.

- b. Find c by integrating with respect to y . (This puts c in the limits of integration.)
- c. Find c by integrating with respect to x . (This puts c into the integrand as well.)
76. Find the area of the region between the curve $y = 3 - x^2$ and the line $y = -1$ by integrating with respect to **a.** x , **b.** y .
77. Find the area of the region in the first quadrant bounded on the left by the y -axis, below by the line $y = x/4$, above left by the curve $y = 1 + \sqrt{x}$, and above right by the curve $y = 2/\sqrt{x}$.
78. Find the area of the region in the first quadrant bounded on the left by the y -axis, below by the curve $x = 2\sqrt{y}$, above left by the curve $x = (y - 1)^2$, and above right by the line $x = 3 - y$.



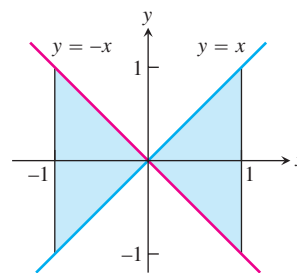
79. The figure here shows triangle AOC inscribed in the region cut from the parabola $y = x^2$ by the line $y = a^2$. Find the limit of the ratio of the area of the triangle to the area of the parabolic region as a approaches zero.



80. Suppose the area of the region between the graph of a positive continuous function f and the x -axis from $x = a$ to $x = b$ is 4 square units. Find the area between the curves $y = f(x)$ and $y = 2f(x)$ from $x = a$ to $x = b$.
81. Which of the following integrals, if either, calculates the area of the shaded region shown here? Give reasons for your answer.

a. $\int_{-1}^1 (x - (-x)) dx = \int_{-1}^1 2x dx$

b. $\int_{-1}^1 (-x - (x)) dx = \int_{-1}^1 -2x dx$



82. True, sometimes true, or never true? The area of the region between the graphs of the continuous functions $y = f(x)$ and $y = g(x)$ and the vertical lines $x = a$ and $x = b$ ($a < b$) is

$$\int_a^b [f(x) - g(x)] dx.$$

Give reasons for your answer.

Theory and Examples

83. Suppose that $F(x)$ is an antiderivative of $f(x) = (\sin x)/x$, $x > 0$. Express

$$\int_1^3 \frac{\sin 2x}{x} dx$$

in terms of F .

84. Show that if f is continuous, then

$$\int_0^1 f(x) dx = \int_0^1 f(1 - x) dx.$$

85. Suppose that

$$\int_0^1 f(x) dx = 3.$$

Find

$$\int_{-1}^0 f(x) dx$$

if **a.** f is odd, **b.** f is even.

86. **a.** Show that if f is odd on $[-a, a]$, then

$$\int_{-a}^a f(x) dx = 0.$$

b. Test the result in part (a) with $f(x) = \sin x$ and $a = \pi/2$.

87. If f is a continuous function, find the value of the integral

$$I = \int_0^a \frac{f(x) dx}{f(x) + f(a - x)}$$

by making the substitution $u = a - x$ and adding the resulting integral to I .

88. By using a substitution, prove that for all positive numbers x and y ,

$$\int_x^{xy} \frac{1}{t} dt = \int_1^y \frac{1}{t} dt.$$

The Shift Property for Definite Integrals

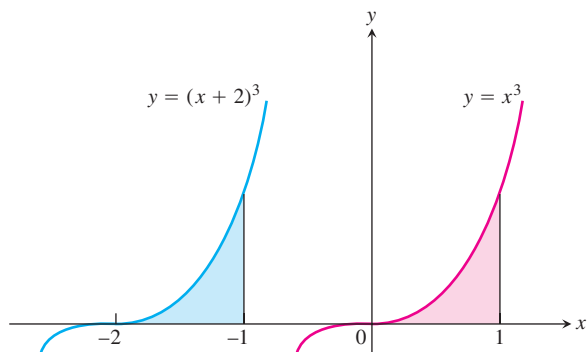
A basic property of definite integrals is their invariance under translation, as expressed by the equation.

$$\int_a^b f(x) dx = \int_{a-c}^{b-c} f(x+c) dx. \quad (1)$$

The equation holds whenever f is integrable and defined for the necessary values of x . For example in the accompanying figure, show that

$$\int_{-2}^{-1} (x+2)^3 dx = \int_0^1 x^3 dx$$

because the areas of the shaded regions are congruent.



89. Use a substitution to verify Equation (1).

90. For each of the following functions, graph $f(x)$ over $[a, b]$ and $f(x+c)$ over $[a-c, b-c]$ to convince yourself that Equation (1) is reasonable.

a. $f(x) = x^2$, $a = 0$, $b = 1$, $c = 1$

b. $f(x) = \sin x$, $a = 0$, $b = \pi$, $c = \pi/2$

c. $f(x) = \sqrt{x-4}$, $a = 4$, $b = 8$, $c = 5$

COMPUTER EXPLORATIONS

In Exercises 91–94, you will find the area between curves in the plane when you cannot find their points of intersection using simple algebra. Use a CAS to perform the following steps:

- Plot the curves together to see what they look like and how many points of intersection they have.
- Use the numerical equation solver in your CAS to find all the points of intersection.
- Integrate $|f(x) - g(x)|$ over consecutive pairs of intersection values.
- Sum together the integrals found in part (c).

91. $f(x) = \frac{x^3}{3} - \frac{x^2}{2} - 2x + \frac{1}{3}$, $g(x) = x - 1$

92. $f(x) = \frac{x^4}{2} - 3x^3 + 10$, $g(x) = 8 - 12x$

93. $f(x) = x + \sin(2x)$, $g(x) = x^3$

94. $f(x) = x^2 \cos x$, $g(x) = x^3 - x$

Chapter 5 Additional and Advanced Exercises

Theory and Examples

1. a. If $\int_0^1 7f(x) dx = 7$, does $\int_0^1 f(x) dx = 1$?

b. If $\int_0^1 f(x) dx = 4$ and $f(x) \geq 0$, does

$$\int_0^1 \sqrt{f(x)} dx = \sqrt{4} = 2?$$

Give reasons for your answers.

2. Suppose $\int_{-2}^2 f(x) dx = 4$, $\int_2^5 f(x) dx = 3$, $\int_{-2}^5 g(x) dx = 2$.

Which, if any, of the following statements are true?

a. $\int_5^2 f(x) dx = -3$

b. $\int_{-2}^5 (f(x) + g(x)) dx = 9$

c. $f(x) \leq g(x)$ on the interval $-2 \leq x \leq 5$

3. **Initial value problem** Show that

$$y = \frac{1}{a} \int_0^x f(t) \sin a(x-t) dt$$

solves the initial value problem

$$\frac{d^2 y}{dx^2} + a^2 y = f(x), \quad \frac{dy}{dx} = 0 \quad \text{and} \quad y = 0 \quad \text{when} \quad x = 0.$$

(Hint: $\sin(ax - at) = \sin ax \cos at - \cos ax \sin at$.)

4. **Proportionality** Suppose that x and y are related by the equation

$$x = \int_0^y \frac{1}{\sqrt{1+4t^2}} dt.$$

Show that $d^2 y/dx^2$ is proportional to y and find the constant of proportionality.

5. Find $f(4)$ if

a. $\int_0^{x^2} f(t) dt = x \cos \pi x$

b. $\int_0^{f(x)} t^2 dt = x \cos \pi x.$

6. Find $f(\pi/2)$ from the following information.

i. f is positive and continuous.

ii. The area under the curve $y = f(x)$ from $x = 0$ to $x = a$ is

$$\frac{a^2}{2} + \frac{a}{2} \sin a + \frac{\pi}{2} \cos a.$$

7. The area of the region in the xy -plane enclosed by the x -axis, the curve $y = f(x)$, $f(x) \geq 0$, and the lines $x = 1$ and $x = b$ is equal to $\sqrt{b^2 + 1} - \sqrt{2}$ for all $b > 1$. Find $f(x)$.

8. Prove that

$$\int_0^x \left(\int_0^u f(t) dt \right) du = \int_0^x f(u)(x - u) du.$$

(Hint: Express the integral on the right-hand side as the difference of two integrals. Then show that both sides of the equation have the same derivative with respect to x .)

9. **Finding a curve** Find the equation for the curve in the xy -plane that passes through the point $(1, -1)$ if its slope at x is always $3x^2 + 2$.
10. **Shoveling dirt** You sling a shovelful of dirt up from the bottom of a hole with an initial velocity of 32 ft/sec. The dirt must rise 17 ft above the release point to clear the edge of the hole. Is that enough speed to get the dirt out, or had you better duck?

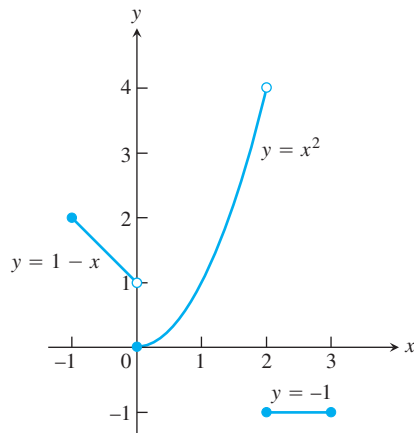


FIGURE 5.34 Piecewise continuous functions like this are integrated piece by piece.

Piecewise Continuous Functions

Although we are mainly interested in continuous functions, many functions in applications are piecewise continuous. A function $f(x)$ is **piecewise continuous on a closed interval I** if f has only finitely many discontinuities in I , the limits

$$\lim_{x \rightarrow c^-} f(x) \text{ and } \lim_{x \rightarrow c^+} f(x)$$

exist and are finite at every interior point of I , and the appropriate one-sided limits exist and are finite at the endpoints of I . All piecewise continuous functions are integrable. The points of discontinuity subdivide I into open and half-open subintervals on which f is continuous, and the limit criteria above guarantee that f has a continuous extension to the closure of each subinterval. To integrate a piecewise continuous function, we integrate the individual extensions and add the results. The integral of

$$f(x) = \begin{cases} 1 - x, & -1 \leq x < 0 \\ x^2, & 0 \leq x < 2 \\ -1, & 2 \leq x \leq 3 \end{cases}$$

(Figure 5.34) over $[-1, 3]$ is

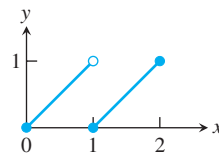
$$\begin{aligned} \int_{-1}^3 f(x) dx &= \int_{-1}^0 (1 - x) dx + \int_0^2 x^2 dx + \int_2^3 (-1) dx \\ &= \left[x - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^3}{3} \right]_0^2 + \left[-x \right]_2^3 \\ &= \frac{3}{2} + \frac{8}{3} - 1 = \frac{19}{6}. \end{aligned}$$

The Fundamental Theorem applies to piecewise continuous functions with the restriction that $(d/dx) \int_a^x f(t) dt$ is expected to equal $f(x)$ only at values of x at which f is continuous. There is a similar restriction on Leibniz's Rule below.

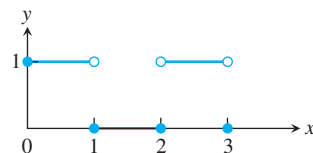
Graph the functions in Exercises 11–16 and integrate them over their domains.

11. $f(x) = \begin{cases} x^{2/3}, & -8 \leq x < 0 \\ -4, & 0 \leq x \leq 3 \end{cases}$
12. $f(x) = \begin{cases} \sqrt{-x}, & -4 \leq x < 0 \\ x^2 - 4, & 0 \leq x \leq 3 \end{cases}$
13. $g(t) = \begin{cases} t, & 0 \leq t < 1 \\ \sin \pi t, & 1 \leq t \leq 2 \end{cases}$
14. $h(z) = \begin{cases} \sqrt{1 - z}, & 0 \leq z < 1 \\ (7z - 6)^{-1/3}, & 1 \leq z \leq 2 \end{cases}$
15. $f(x) = \begin{cases} 1, & -2 \leq x < -1 \\ 1 - x^2, & -1 \leq x < 1 \\ 2, & 1 \leq x \leq 2 \end{cases}$
16. $h(r) = \begin{cases} r, & -1 \leq r < 0 \\ 1 - r^2, & 0 \leq r < 1 \\ 1, & 1 \leq r \leq 2 \end{cases}$

17. Find the average value of the function graphed in the accompanying figure.



18. Find the average value of the function graphed in the accompanying figure.



Leibniz's Rule

In applications, we sometimes encounter functions like

$$f(x) = \int_{\sin x}^{x^2} (1+t) dt \quad \text{and} \quad g(x) = \int_{\sqrt{x}}^{2\sqrt{x}} \sin t^2 dt,$$

defined by integrals that have variable upper limits of integration and variable lower limits of integration at the same time. The first integral can be evaluated directly, but the second cannot. We may find the derivative of either integral, however, by a formula called **Leibniz's Rule**.

Leibniz's Rule

If f is continuous on $[a, b]$ and if $u(x)$ and $v(x)$ are differentiable functions of x whose values lie in $[a, b]$, then

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx}.$$

Figure 5.35 gives a geometric interpretation of Leibniz's Rule. It shows a carpet of variable width $f(t)$ that is being rolled up at the left at the same time x as it is being unrolled at the right. (In this interpretation, time is x , not t .) At time x , the floor is covered from $u(x)$ to $v(x)$. The rate du/dx at which the carpet is being rolled up need not be the same as the rate dv/dx at which the carpet is being laid down. At any given time x , the area covered by carpet is

$$A(x) = \int_{u(x)}^{v(x)} f(t) dt.$$

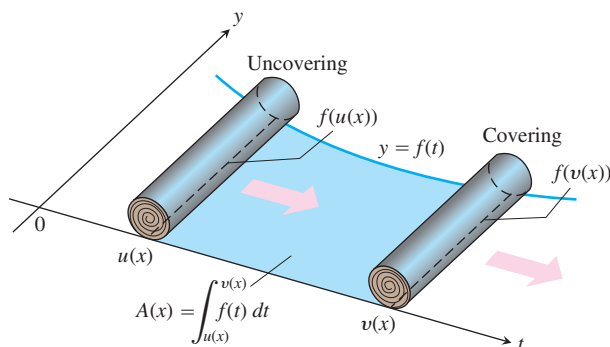


FIGURE 5.35 Rolling and unrolling a carpet: a geometric interpretation of Leibniz's Rule:

$$\frac{dA}{dx} = f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx}.$$

At what rate is the covered area changing? At the instant x , $A(x)$ is increasing by the width $f(v(x))$ of the unrolling carpet times the rate

dv/dx at which the carpet is being unrolled. That is, $A(x)$ is being increased at the rate

$$f(v(x)) \frac{dv}{dx}.$$

At the same time, A is being decreased at the rate

$$f(u(x)) \frac{du}{dx},$$

the width at the end that is being rolled up times the rate du/dx . The net rate of change in A is

$$\frac{dA}{dx} = f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx},$$

which is precisely Leibniz's Rule.

To prove the rule, let F be an antiderivative of f on $[a, b]$. Then

$$\int_{u(x)}^{v(x)} f(t) dt = F(v(x)) - F(u(x)).$$

Differentiating both sides of this equation with respect to x gives the equation we want:

$$\begin{aligned} \frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt &= \frac{d}{dx} [F(v(x)) - F(u(x))] \\ &= F'(v(x)) \frac{dv}{dx} - F'(u(x)) \frac{du}{dx} \quad \text{Chain Rule} \\ &= f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx}. \end{aligned}$$

Use Leibniz's Rule to find the derivatives of the functions in Exercises 19–21.

$$19. f(x) = \int_{1/x}^x \frac{1}{t} dt \quad 20. f(x) = \int_{\cos x}^{\sin x} \frac{1}{1-t^2} dt$$

$$21. g(y) = \int_{\sqrt{y}}^{2\sqrt{y}} \sin t^2 dt$$

22. Use Leibniz's Rule to find the value of x that maximizes the value of the integral

$$\int_x^{x+3} t(5-t) dt.$$

Problems like this arise in the mathematical theory of political elections. See “The Entry Problem in a Political Race,” by Steven J. Brams and Philip D. Straffin, Jr., in *Political Equilibrium*, Peter Ordeshook and Kenneth Shephle, Editors, Kluwer-Nijhoff, Boston, 1982, pp. 181–195.

Approximating Finite Sums with Integrals

In many applications of calculus, integrals are used to approximate finite sums—the reverse of the usual procedure of using finite sums to approximate integrals.